

$$F(s) = L\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt$$

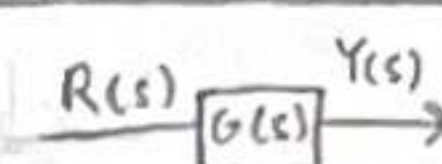
Euler's Formula:
 $e^{j\theta} = \cos\theta + j\sin\theta$
 $e^{-j\theta} = \cos\theta - j\sin\theta$

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$\sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$\text{sinc}(\theta) = \frac{\sin(\pi\theta)}{\pi\theta}$$

Transfer Function:



$$Y(s) = G(s) \cdot R(s)$$

Laplace Transform:

Properties:

$$f(t) \longleftrightarrow F(s)$$

$$\delta(t) \longleftrightarrow 1$$

$$u(t) \longleftrightarrow \frac{1}{s}$$

$$t \longleftrightarrow \frac{1}{s^2}$$

① Time Delay, $f(t-T) \longleftrightarrow e^{-Ts} F(s)$

② Differentiation,

$$f'(t) \longleftrightarrow sF(s) - f(0)$$

$$s^2 F(s) - s f(0) - f'(0) \longleftrightarrow s^2 F(s) - f(0) + F(s)$$

③ Final Value Thm, "if all the poles of $sF(s)$ are in the LHP with possibly one simple pole at the origin"

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s F(s)$$

④ Initial Value Thm, "if the limit exists, i.e. no requirements"

$$\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} s F(s)$$

⑤ Frequency shift,

$$e^{-at} f(t) \longleftrightarrow F(s+a)$$

$$t^n \longleftrightarrow \frac{n!}{s^{n+1}}$$

$$e^{-at} \longleftrightarrow \frac{1}{s+a}$$

$$\sin(at) \longleftrightarrow \frac{a}{s^2 + a^2}$$

$$\cos(at) \longleftrightarrow \frac{s}{s^2 + a^2}$$

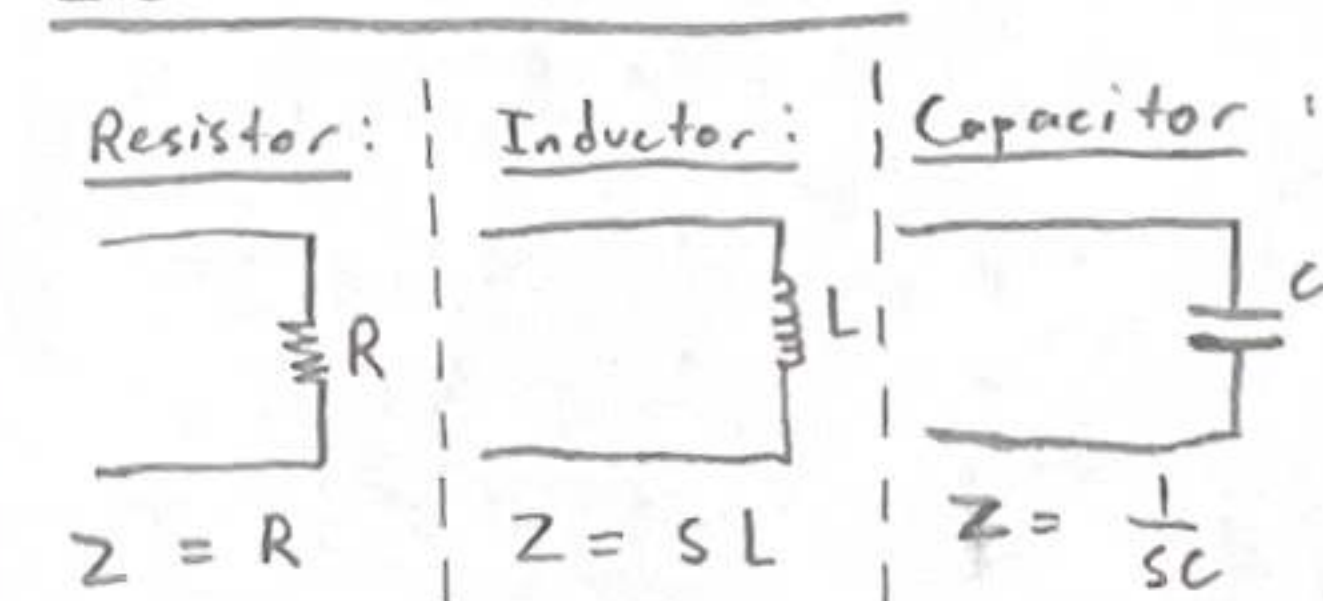
$$te^{-at} \longleftrightarrow \frac{1}{(s+a)^2}$$

$$e^{-at} \sin(bt) \longleftrightarrow \frac{b}{(s+a)^2 + b^2}$$

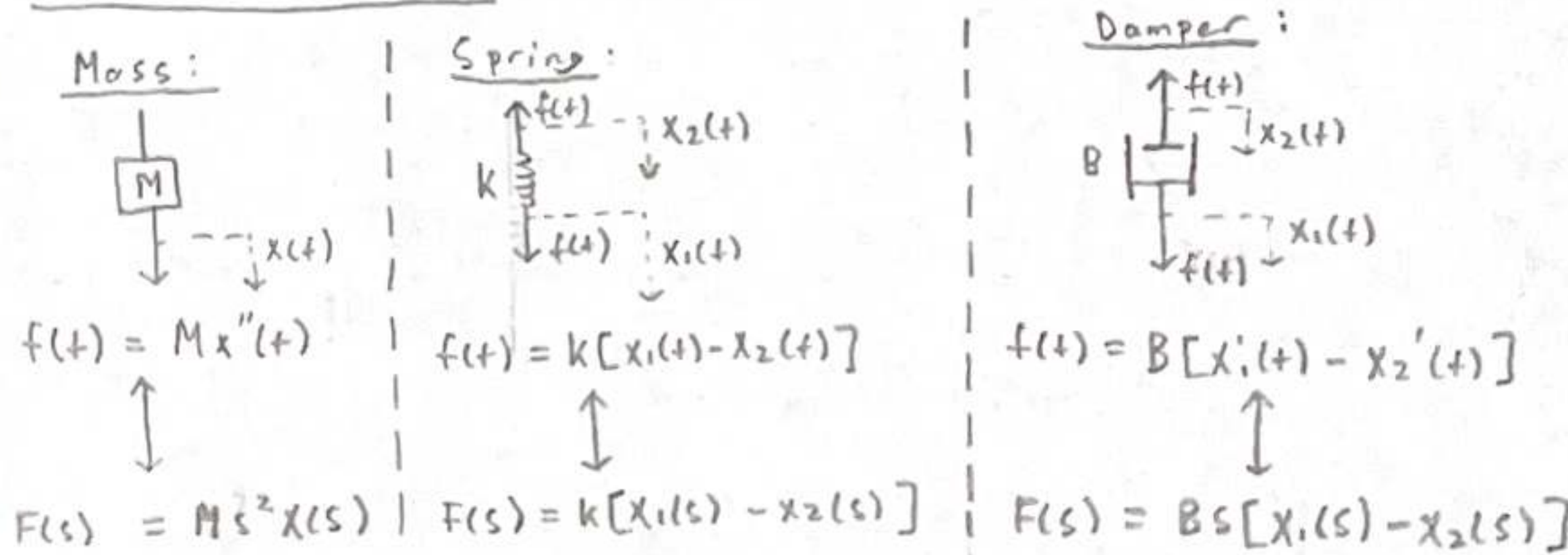
$$e^{-at} \cos(bt) \longleftrightarrow \frac{s+a}{(s+a)^2 + b^2}$$

$$t^n f(t) \longleftrightarrow (-1)^n \frac{d}{ds^n} (F(s))$$

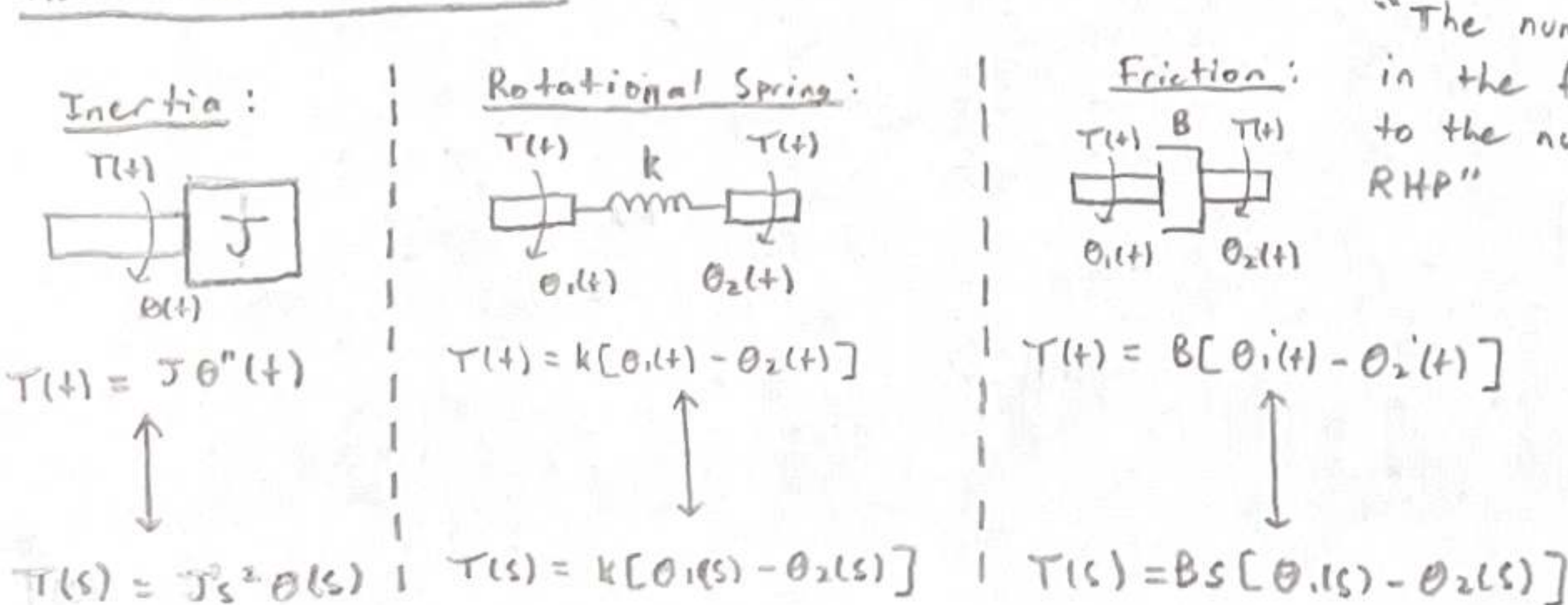
Electrical Elements:



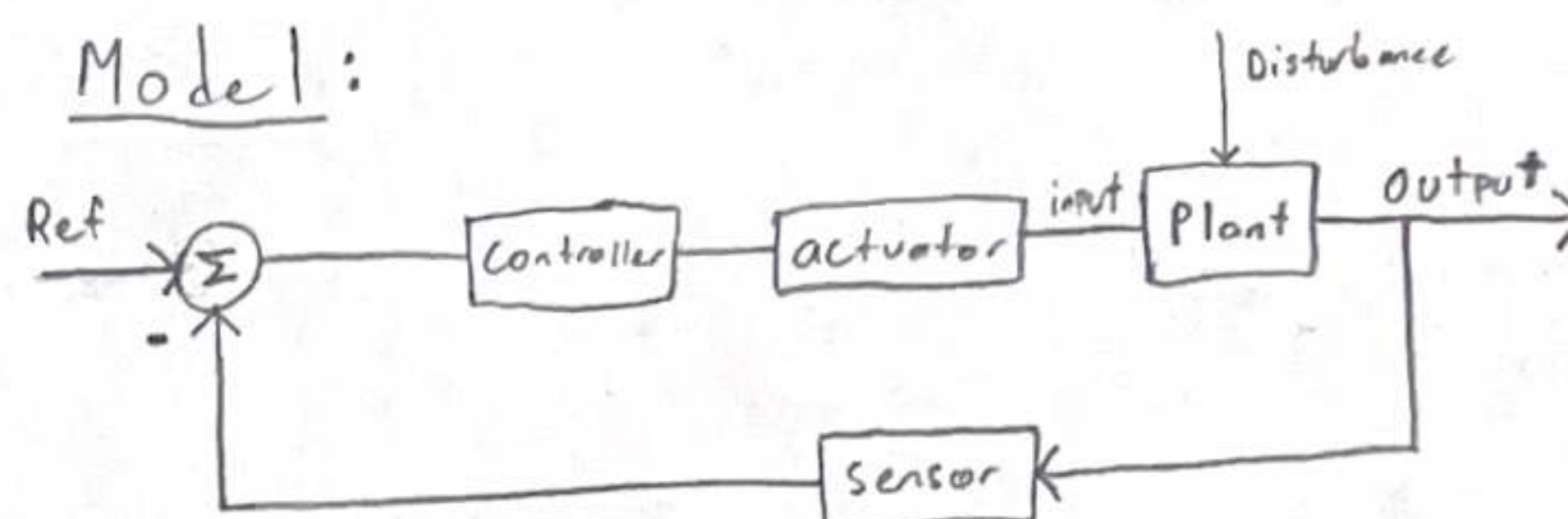
Mechanical Elements:



Rotational Elements:



Model:

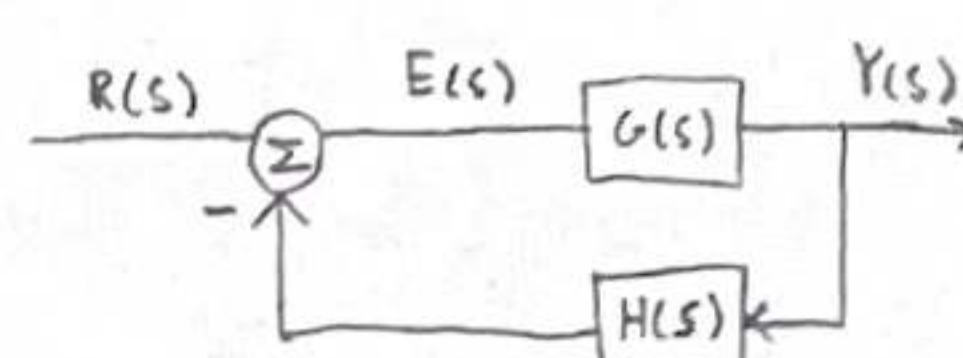


Similarities:

- ① Capacitor ~ Mass ~ Inertia
- ② Inductor ~ Spring ~ Rotational Spring
- ③ Resistor ~ Damper ~ Friction

Definitions:

- ① $G(s)$ = Forward Transfer Function
- ② $H(s)$ = Feedback Transfer Function
- ③ $G(s)H(s)$ = open-loop TF
- ④ $\frac{Y(s)}{R(s)}$ = closed-loop TF
- ⑤ $\frac{Y(s)}{E(s)}$ = Feed-forward TF



Black's Formula:

$$Y(s) = \frac{G(s)}{1 + G(s)H(s)} \cdot R(s)$$

* if Positive Feedback, $\oplus \rightarrow \ominus$

Tricks:

- ① 1st Order, all coefficients have the same sign then all poles in LHP
- ② 2nd order, all coefficients have the same sign then all poles in LHP
- ③ 3rd order, if even one coefficient has a different sign you will have roots in RHP

Stability:

- ① STABLE $\rightarrow$ All poles in LHP
- ② Marginally STABLE
 - No poles in RHP
 - All poles on $j\omega$ axis have multiplicity 1
- ③ UNSTABLE $\rightarrow$ Poles in RHP

Routh-Array:

s^4	a_n	a_{n-2}	a_{n-4}	$b_1 = \frac{(a_{n-1})(a_{n-2}) - (a_n)(a_{n-3})}{(a_{n-1})}$
s^3	a_{n-1}	a_{n-3}		$b_2 = \frac{(a_{n-1})(a_{n-4}) - (a_n)(a_{n-5})}{(a_{n-1})}$
s^2	b_1	b_2		$c_1 = \frac{(b_1)(a_{n-3}) - (a_{n-1})(b_2)}{(b_1)}$
s^1	c_1			$c_2 = \frac{(b_1)(a_{n-5}) - (a_{n-1})(b_2)}{(b_1)}$
s^0	d_1			

"The number of sign changes in the first column is equal to the number of roots in the RHP"

Ex: Inverse of a 3x3 Matrix

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 2 & -1 & 3 \\ 1 & 1 & 4 \end{bmatrix} \det(A) = 3$$

$$\begin{bmatrix} +|-1 & 3| & -|2 & 3| & +|2 & -1| \\ -|0 & 1| & +|0 & 1| & -|0 & 0| \\ +|0 & 1| & -|0 & 1| & +|0 & 0| \end{bmatrix}$$

$$= \begin{bmatrix} -7 & -5 & 3 \\ 1 & -1 & 0 \\ 1 & 2 & 0 \end{bmatrix} \text{ * Flip over Diagonal}$$

Ex: Inverse of a 2x2

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

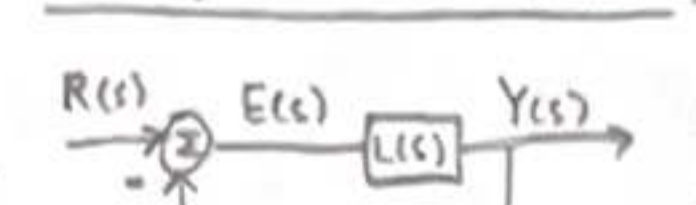
Time Response

$$y(t) = y_{tr}(t) + y_{ss}(t)$$

Transient Steady-state

- ① Transient = 0 as $t \rightarrow \infty$
- ② Steady-state = $\lim_{t \rightarrow \infty} y(t)$

Steady-state Error



Unity feedback = 1 on the bottom

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \cdot \frac{1}{1+L(s)} \cdot R(s)$$

Error Constants:

Step-Error: $k_p = \lim_{s \rightarrow 0} L(s)$

Ramp-Error: $k_v = \lim_{s \rightarrow 0} s \cdot L(s)$

Parabolic-Error: $k_a = \lim_{s \rightarrow 0} s^2 \cdot L(s)$

ess for Inputs:

① $r(t) = R u(t)$ $e_{ss} = \frac{R}{1+k_p}$

② $r(t) = R + u(t)$ $e_{ss} = \frac{R}{k_v}$

③ $r(t) = \frac{R t^2}{2} u(t)$ $e_{ss} = \frac{R}{k_a}$

Accurate Tracking ($e_{ss} = 0$):

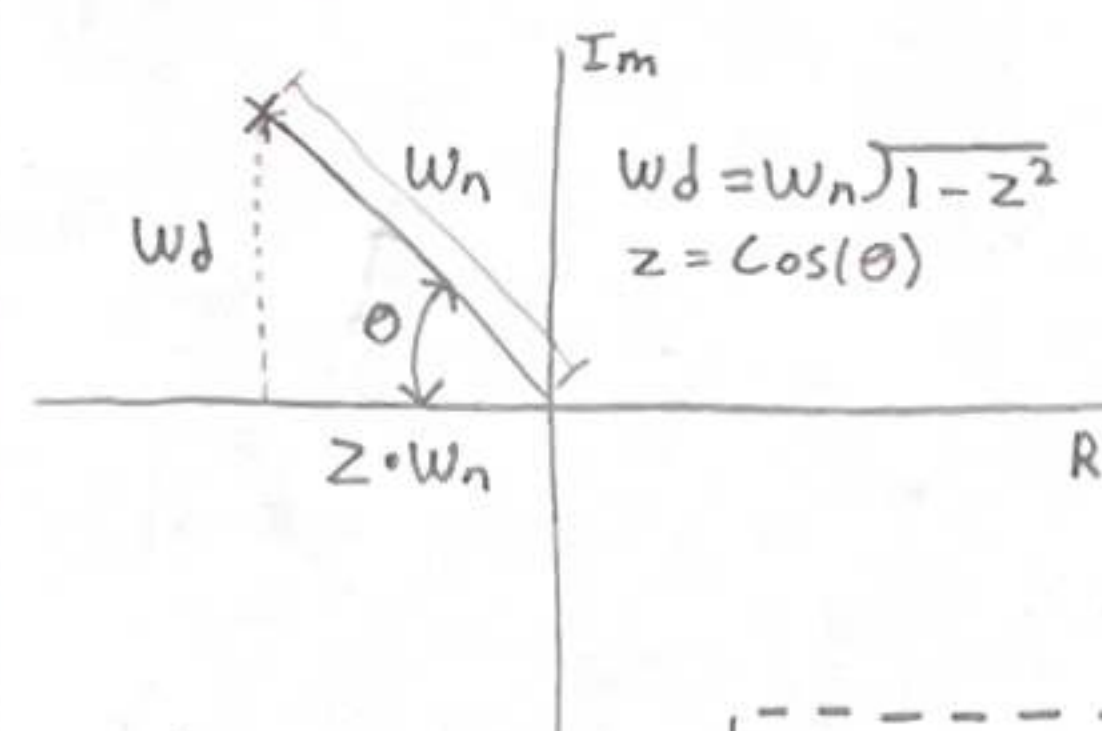
$$k_p = k_v = k_a = \infty$$

Time Values Graphically:

- ① Delay time, time to reach 0.5 y_{ss}
- ② Rise time, 0.1 y_{ss} - 0.9 y_{ss}
- ③ Settling time, time when it enters percent box and stays in it
- ④ Percent overshoot, $\frac{y_{max} - y_{ss}}{y_{ss}} \cdot 100\%$
- ⑤ Peak Time, time to reach y_{max}

Location of Pole gives you Everything:

[2nd Order] *0 < z < 1*



1st Order System:

$$G(s) = \frac{k}{Ts+1}$$

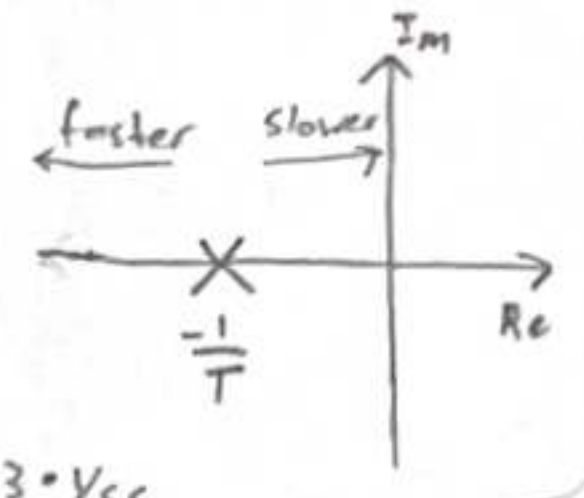
- ① DC Gain, $\lim_{t \rightarrow \infty} y(t) = k$
- ② Time Constant, T when @ 0.63 y_{ss}

For step response:

- ① $y_{ss} = k$
- ② $y_{max}, T_p, PO = \text{UNDEFINED}$
- ③ Delay time = 0.7 T
- ④ Rise time = 2.2 T
- ⑤ Settling time, 20% = 4T, 5% = 3T

Pole Locations:

- ① Undamped, Poles on jw axis
- ② Underdamped, Distinct Complex Poles
- ③ Critically damped, Double Real Poles
- ④ Overdamped, Distinct Real Poles



2nd Order System:

$$G(s) = \frac{k}{s^2 + 2zW_n s + W_n^2}$$

- ① Damping Ratio, 'z'
- ② Undamped Natural Frequency, 'Wn'

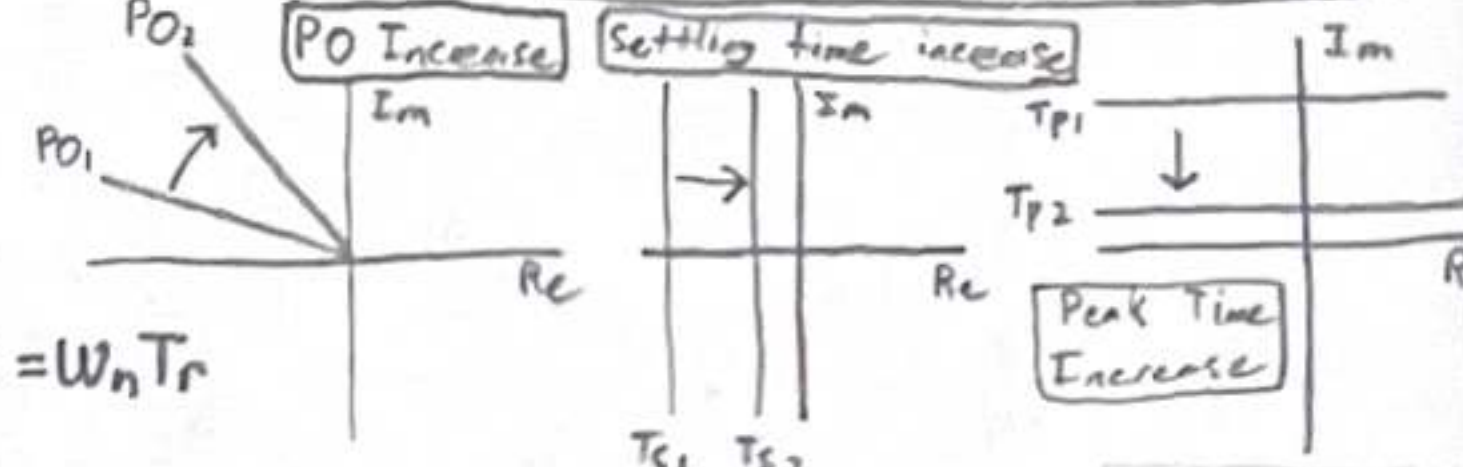
Properties of 2nd Order Systems:

- ① Settling time, 20% = $\frac{4}{z \cdot W_n}$, 5% = $\frac{3}{z \cdot W_n}$
- ② Peak time, $\frac{\pi}{W_d}$
- ③ Peak Value, $1 + e^{-2 \cdot \pi / \sqrt{1-z^2}}$
- ④ Percent overshoot, 100e
- ⑤ Rise Time, $1.76z^3 - 0.417z^2 + 1.039z + 1 = W_n T_r$
- ⑥ $y_{max} = 1 + e^{-\frac{2 \cdot \pi}{\sqrt{1-z^2}}}$

Damping Ratios:

- ① Undamped, z = 0
- ② Underdamped, 0 < z < 1
- ③ Critically Damped, z = 1
- ④ Overdamped, z > 1

How to increase Values:



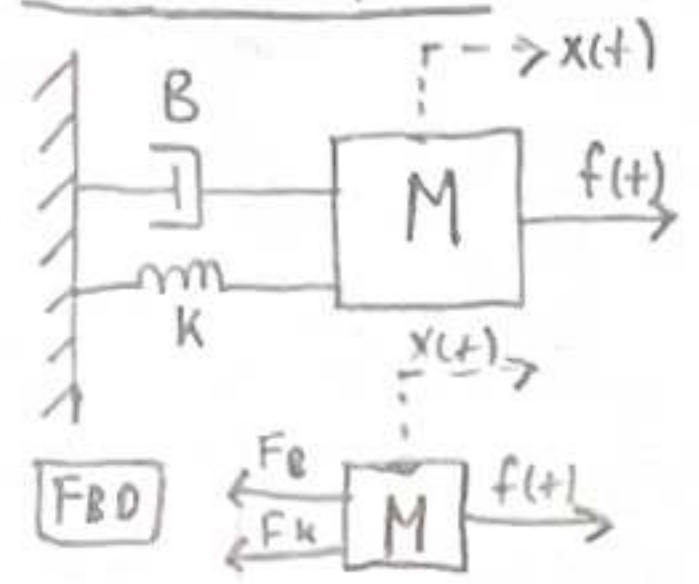
Complex Formula:

$$z = \frac{\ln(\frac{PO}{100})}{\sqrt{\pi^2 + (\ln(\frac{PO}{100}))^2}}$$

Dominant Poles

$$G(s) = \frac{10}{(s+1)(s+10)} \approx \frac{1}{s+1} \quad (s+10 \text{ Far away})$$

FBD Example:



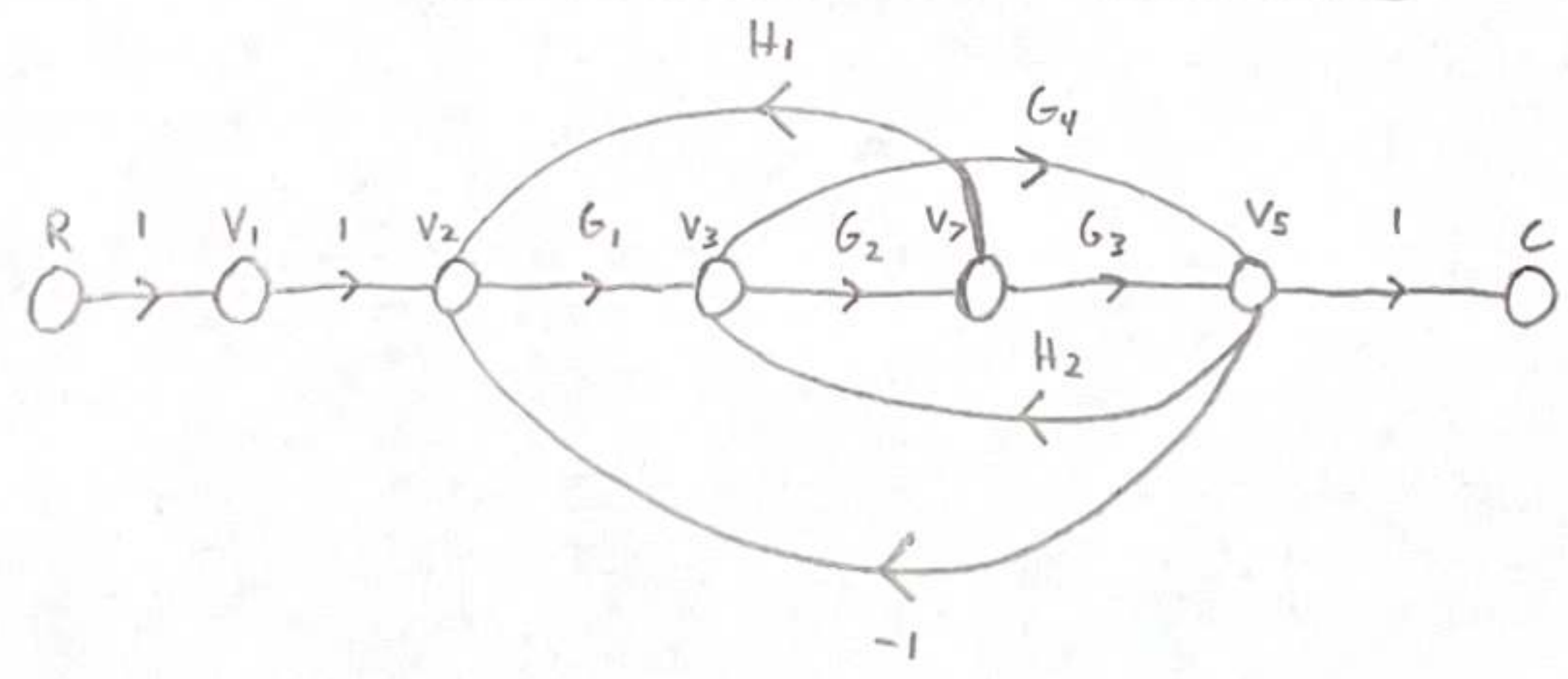
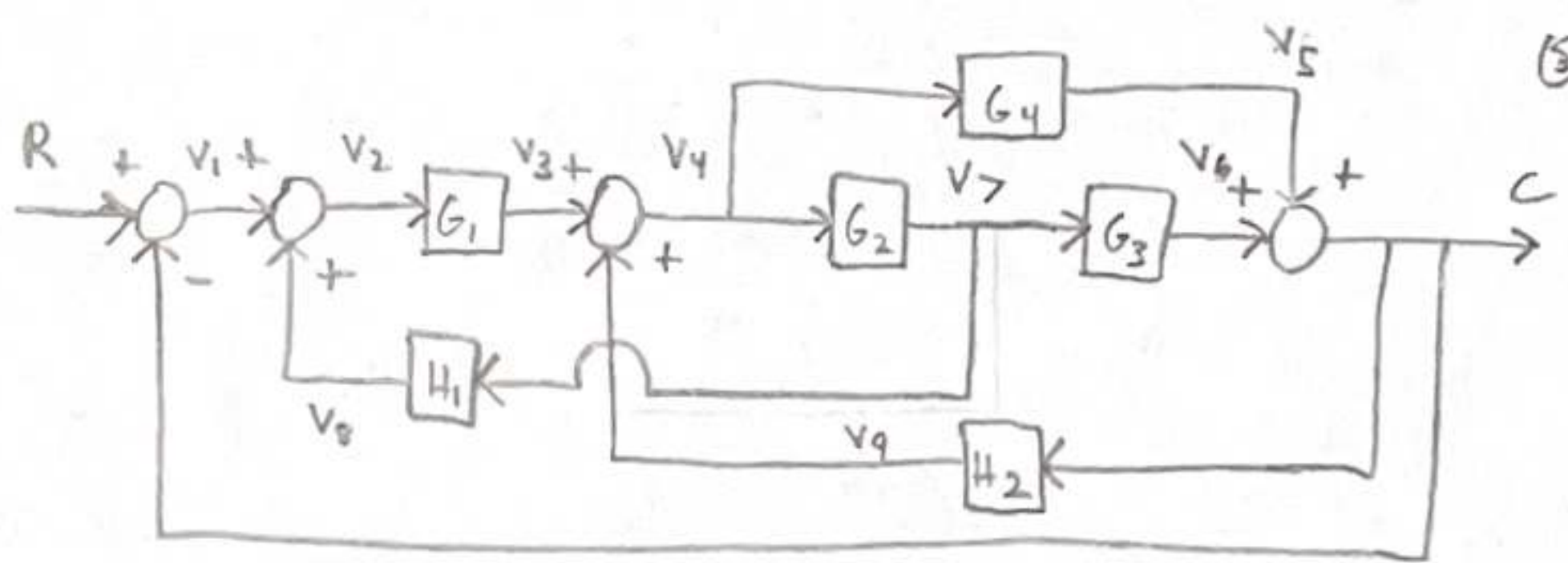
$$M\ddot{x} = f(t) - F_B - F_k$$

$$\frac{X(s)}{F(s)} = \frac{1}{Ms^2 + Bs + k}$$

$$F_B = (-B)(\dot{x}_{LEFT} - \dot{x}_{RIGHT})$$

$$F_k = (-K)(x_{LEFT} - x_{RIGHT})$$

Signal Flow Example:



Ex: "Linearize"

$$m\ddot{v}(t) + k_f v(t) + k_a v^2(t) = F_m(t) \quad @ OP \rightarrow V_o = 30$$

- ① Input: $V(t)$ output: $F_m(t)$
- ② Operating Point: $V_o = 30 \rightarrow V(t) = V_o \rightarrow \dot{V}(t) = 0$
 $\therefore m\ddot{v}(t) + k_f v(t) + k_a v^2(t) = F_m(t)$
 $k_f V_o + k_a V_o^2 = F_{m0}$

Taylor Series Expansion:

$$f(\dot{v}, v, F_m) = m\ddot{v}(t) + k_f v(t) + k_a v^2(t) - F_m(t)$$

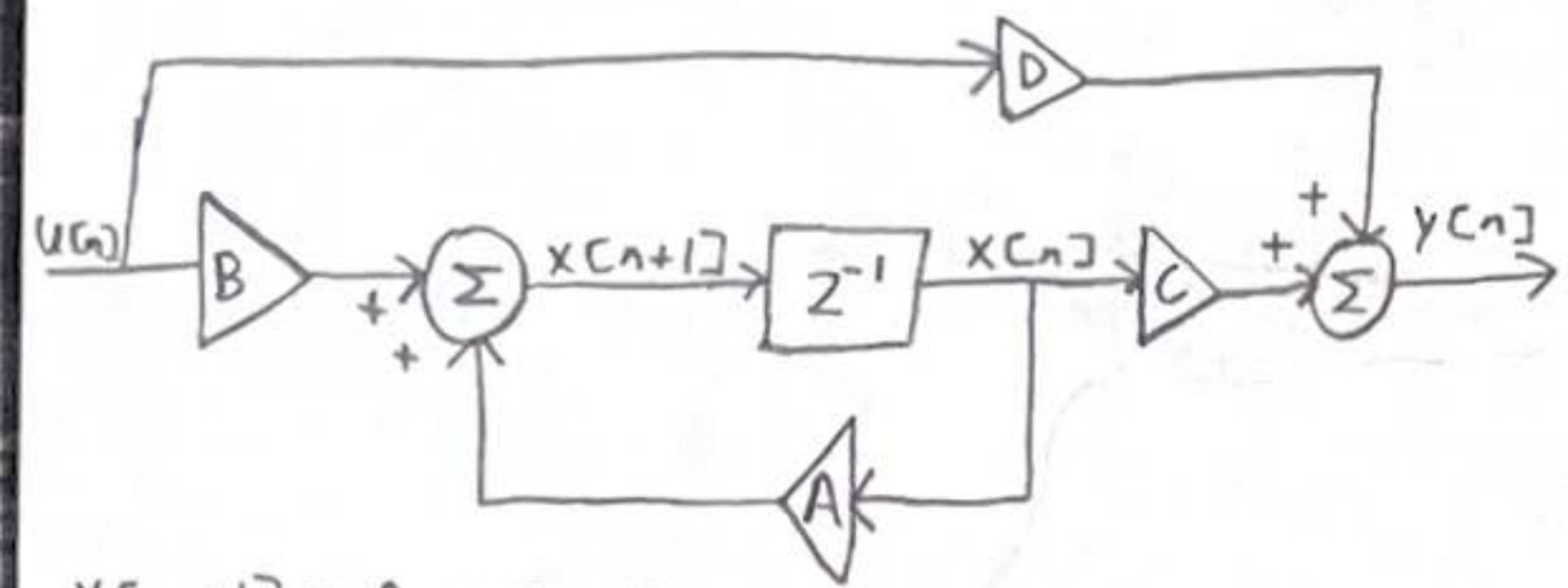
$$\frac{df}{dv} \bigg|_{op} = m, \quad \frac{df}{dv} \bigg|_{op} = k_f + 2k_a V_o, \quad \frac{df}{dF_m} \bigg|_{op} = -1$$

$$\therefore f(\dot{v}, v, F_m) = f(\dot{V}_o, V_o, F_{m0}) + m(\dot{v} - \dot{V}_o) + (k_f + 2k_a V_o)(v - V_o) - (F_m - F_{m0})$$

$$0 = 0 + m(\dot{v} - \dot{V}_o) + (k_f + 2k_a V_o)(v - V_o) - (F_m - F_{m0})$$

$$\therefore F_m - F_{m0} = m(\dot{v} - \dot{V}_o) + (k_f + 2k_a V_o)(v - V_o)$$

$$F_m - F_{m0} = m(\dot{v} - \dot{V}_o) + (k_f + 2k_a V_o)(v - V_o)$$



$$x[n+1] = Ax[n] + Bu[n]$$

$$y[n] = Cx[n] + Du[n]$$

State Transition Matrix

$$x[n] = \underbrace{A^n x_0}_{\text{ZIR}} + \underbrace{\sum_{k=0}^{n-1} A^{n-1-k} Bu[k]}_{\text{ZSR}}$$

$$y[n] = \underbrace{CA^n x_0}_{\text{ZIR}} + \underbrace{\sum_{k=0}^{n-1} CA^{n-1-k} Bu[k]}_{\text{ZSR}} + Du[n]$$

$$Y(z) = C(zI - A)^{-1} x_0 + [C(zI - A)^{-1} B + D] U(z)$$

$H(z)$

Ex: "DT Fourier Transform"

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= \sum_{n=-18}^{18} 8 e^{-j\omega n} \rightarrow \frac{8(1 - e^{-j\omega(19)})}{1 - e^{-j\omega}}$$

$$= 8e^{-j\omega(19/2)} \frac{[e^{j\omega(19/2)} - e^{-j\omega(19/2)}]}{[e^{j\omega(1/2)} - e^{-j\omega(1/2)}]}$$

$$= 8e^{-j\omega(19/2)} \cdot \frac{\sin(19\omega/2)}{\sin(\omega/2)}$$

How to Compute State Transition Matrix?

① $\det(A - \lambda I) = 0 \rightarrow Av_i = \lambda_i v_i$

$$A = [v_1, v_2, \dots, v_n] \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix} [v_1, v_2, \dots, v_n]^{-1} = TDT^{-1}$$

② Cayley-Hamilton Theorem

State Transition matrix

$$\xrightarrow{CT} e^{At} = T \begin{bmatrix} e^{\lambda_1 t} & 0 & \dots & 0 \\ 0 & e^{\lambda_2 t} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{\lambda_n t} \end{bmatrix} T^{-1}$$

$$\xrightarrow{DT} A^n = T \begin{bmatrix} \lambda_1^n & 0 & \dots & 0 \\ 0 & \lambda_2^n & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n^n \end{bmatrix} T^{-1}$$

State Transition matrix

$$\xrightarrow{CT} e^{At} = \sum_{k=0}^{N-1} C_k A^k \text{ where } v_i \in [1, N]$$

$$e^{\lambda_i t} = \sum_{k=0}^{N-1} C_k \lambda_i^k$$

$$\xrightarrow{DT} A^n = \sum_{k=0}^{N-1} C_k A^k \text{ where } v_i \in [1, N]$$

$$\lambda_i^n = \sum_{k=0}^{N-1} C_k \lambda_i^k$$

Frequency Response

$$x(t) = e^{j\omega t} \xrightarrow{CT} y(t) = |H(\omega)| e^{j(\omega t + \angle H(\omega))}$$

$$x[n] = e^{j\omega n} \xrightarrow{DT} y[n] = |H(\omega)| e^{j(\omega n + \angle H(\omega))}$$

Ex: "Matrix Multiplication"

$$A = \begin{bmatrix} 3 & -1 & 0 \\ 2 & 5 & 1 \\ -7 & 1 & 3 \end{bmatrix} B = \begin{bmatrix} 6 & -1 & 0 \\ 0 & 1 & -2 \\ 3 & -8 & 1 \end{bmatrix}$$

Inverse of a 2x2 Matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$AB = \begin{bmatrix} (3)(6) + (-1)(0) + (0)(3) & (3)(-1) + (-1)(1) + (0)(-8) & (3)(0) + (-1)(-2) + (0)(1) \\ (2)(6) + (5)(0) + (1)(3) & (2)(-1) + (5)(1) + (1)(-8) & (2)(0) + (5)(-2) + (1)(1) \\ (-7)(6) + (1)(0) + (3)(3) & (-7)(-1) + (1)(1) + (3)(-8) & (-7)(0) + (1)(-2) + (3)(1) \end{bmatrix}$$

Ex: "Inverse of a 3x3 Matrix"

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 2 & -1 & 3 \\ 1 & 1 & 4 \end{bmatrix} \det(A) = 3$$

$$A^{-1} = \frac{1}{3} \begin{bmatrix} -7 & 1 & 1 \\ -5 & -1 & 2 \\ 3 & 0 & 0 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} +|-13| & -|23| & +|2-1| \\ -|01| & +|01| & -|00| \\ +|01| & -|01| & +|00| \end{bmatrix}$$

$$= \begin{bmatrix} -7 & -5 & 3 \\ 1 & -1 & 0 \\ 1 & 2 & 0 \end{bmatrix} \text{ * Flip over diagonal}$$

Ex: "Cayley Hamilton Theorem"

$$A = \begin{bmatrix} 0 & 7 \\ -22 & 17 \end{bmatrix} \rightarrow \lambda^2 - 17\lambda + 72 = 0$$

$$(\lambda - 9)(\lambda - 8) \therefore \lambda_1 = 8, \lambda_2 = 9$$

$$e^{At} = A^n = \alpha_0 I + \alpha_1 A$$

$$f(A) = \alpha_0 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \alpha_1 \begin{bmatrix} 0 & 7 \\ -22 & 17 \end{bmatrix}$$

For Bigger Matrices:

$$e^{At} = \alpha_0 I + \alpha_1 A + \alpha_2 A^2$$

$$A^n = \alpha_0 I + \alpha_1 A + \alpha_2 A^2$$

Transfer function from State Diagram

$$H(s) = \frac{b_0 s^N + b_1 s^{N-1} + \dots + b_N}{s^N + a_1 s^{N-1} + \dots + a_N}$$

Ex: "Differentiation Property"

$$x(t) = \begin{cases} 0, & t < -1/2 \\ +1/2, & -1/2 \leq t \leq 1/2 \\ 1, & t > 1/2 \end{cases}$$

$$x(t) = r(t + 1/2) - r(t - 1/2)$$

$$\frac{dx(t)}{dt} = u(t + 1/2) - u(t - 1/2)$$

$$\frac{d^2 x(t)}{dt^2} = \delta(t + 1/2) - \delta(t - 1/2)$$

$\uparrow$ FT

$$(j\omega)^2 X(\omega) = e^{j(\omega/2)} - e^{-j(\omega/2)}$$

$$X(\omega) = \frac{2j \sin(\omega/2)}{(j\omega)^2}$$

$$X(\omega) = \frac{2 \sin(\omega/2)}{j\omega^2}$$

Ex: "DT Response"

"Consider a filter with difference equation $y[n] - \frac{1}{4}y[n-2] = x[n] + \frac{1}{2}x[n-1]$ "

SOLUTION:

$$Y(z) - \frac{1}{4}z^{-2}Y(z) = X(z) + \frac{1}{2}z^{-1}X(z)$$

$$Y(z)[1 - \frac{1}{4}z^{-2}] = X(z)[1 + \frac{1}{2}z^{-1}]$$

$$H(z) = \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{1}{4}z^{-2}} = \frac{1}{1 - \frac{1}{2}z^{-1}}$$

- Find
- Frequency and impulse response
 - Output when input is $x[n] = \cos(\frac{\pi n}{2})$

FREQ Response

$$H(\omega) = \frac{1}{1 - \frac{1}{2}e^{-j\omega}}$$

IMPULSE RESPONSE

$$h[n] = z^{-1}\{H(z)\} = (\frac{1}{2})^n u[n]$$

$$H(\frac{\pi}{2}) = \frac{1}{1 - \frac{1}{2}e^{-j(\pi/2)}} = \frac{1}{1 + \frac{1}{2}j} \leftarrow z_1$$

$$|H(\frac{\pi}{2})| = \frac{1}{\sqrt{1 + (\frac{1}{2})^2}} \quad \angle H(\frac{\pi}{2}) = 0 - \tan^{-1}(\frac{1/2}{1})$$

DT Causalities

- Causal, attach $u[n]$ ie: $(A[n] + B[n])u[n]$
- Two-sided, attach both ie: $A[n]u[n] + B[n](-u[-n-1])$
- Anti-causal, attach $(-u[-n-1])$ ie: $(A[n] + B[n])(-u[-n-1])$

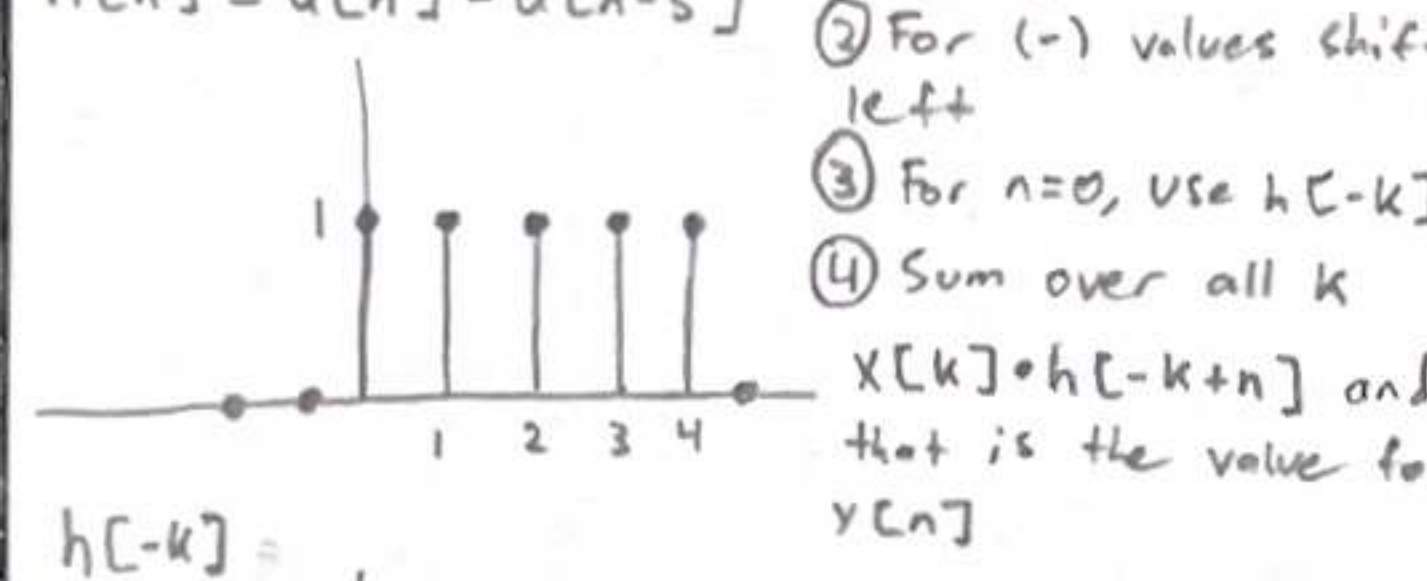
Ex: "Discrete Time Convolution"

The impulse response of a discrete LTI system is given by $h[n] = u[n] - u[n-5]$. Given that the input to the system is given by $x[n] = 3(u[n] - u[n-6])$. Find $y[n]$.

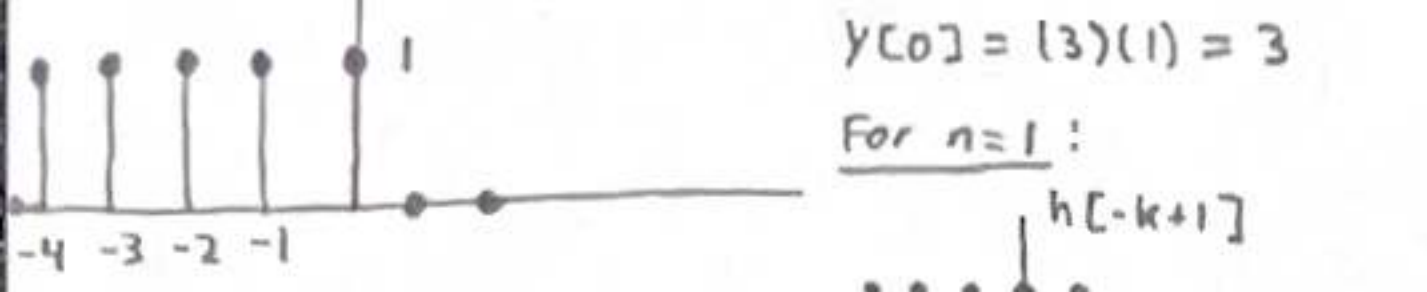
SOLUTION:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

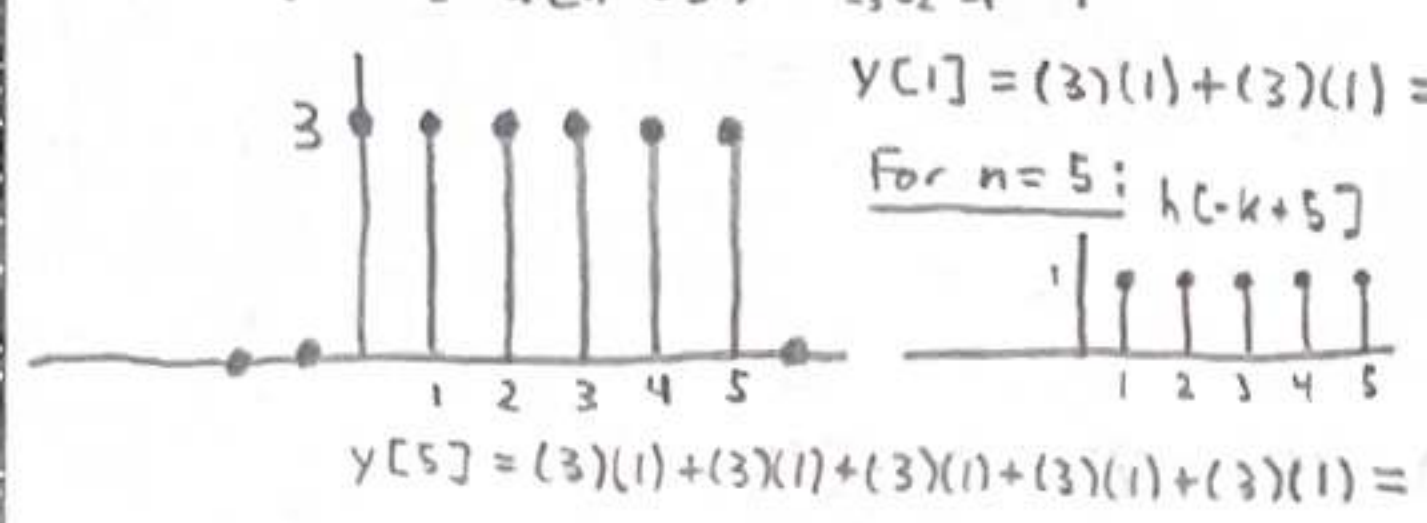
$$h[n] = u[n] - u[n-5]$$



$$h[-k] =$$



$$x[n] = 3(u[n] - u[n-6])$$

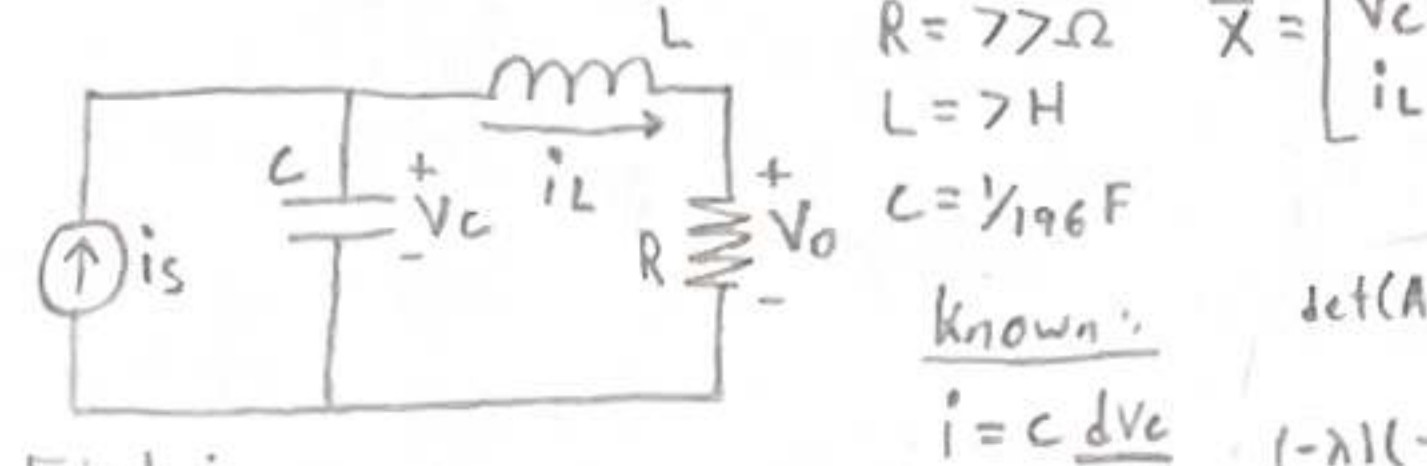


$$y[5] = (3)(1) + (3)(1) + (3)(1) + (3)(1) + (3)(1) = 15$$

Ex: "Different Causalities"

Impulse Response:	Laplace Transform:	ROC:	BIBO:
$8e^{-6t}u(t) + 15e^{2t}u(t)$	$\frac{8}{s+6} + \frac{15}{s-2}$	$(2, \infty)$	NO
$-7e^{-4t}u(-t) - 15e^{2t}u(-t)$	$\frac{-7}{(-s+4)} - \frac{15}{(-s+2)}$	$(-\infty, -4)$	NO
$19e^{-6t}u(t) - 13e^{2t}u(-t)$	$\frac{19}{s+6} - \frac{13}{(-s+2)}$	$(-6, 2)$	YES

Ex: "State Space Circuit"



Find:

- State space Representation $[A, B, C, D]$
- Transfer function $H(s)$
- State Transition Matrix $\Phi(t)$
- Time domain response with zero-input and initial conditions $x_1(0)=4, x_2(0)=5$

SOLUTION:

$$KCL: i_s = C \frac{dV_c}{dt} + i_L \quad KVL: V_c - L \frac{di_L}{dt} - R \cdot i_L = 0$$

$$C \frac{dV_c}{dt} = i_s - i_L \quad L \frac{di_L}{dt} = V_c - R \cdot i_L$$

$$\dot{V}_c = \frac{1}{C} i_s - \frac{1}{C} i_L \quad \dot{i}_L = \frac{1}{L} V_c - \frac{R}{L} i_L$$

$$KCL: \frac{V_o}{R} = i_L \rightarrow V_o = R \cdot i_L$$

$$\begin{bmatrix} \dot{V}_c \\ \dot{i}_L \end{bmatrix} = \begin{bmatrix} 0 & -1/C \\ 1/L & -R/L \end{bmatrix} \begin{bmatrix} V_c \\ i_L \end{bmatrix} + \begin{bmatrix} 1/C \\ 0 \end{bmatrix} i_s$$

$$\begin{bmatrix} V_o \end{bmatrix} = \begin{bmatrix} 0 & R \end{bmatrix} \begin{bmatrix} V_c \\ i_L \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} i_s \quad \frac{V_o}{i_s} = \frac{R}{CLs^2 + CRs + 1}$$

Ex: "Aliasing" "A continuous-time signal

$x(t) = 7\cos(23\pi t) + 7\sin(24\pi t) + 4\cos(46\pi t + \pi/9)$ is sampled at 24Hz. Determine $w(t)$ reconstructed using an ideal interpolator at a sampling rate of 1/24s.

SOLUTION:

$$f_s = 24 \quad \therefore \text{Reconstructed sine completely} \quad \therefore \text{DC value because}$$

$$W_s = 48\pi \quad \text{③ } 4\cos(46\pi t + \pi/9) \quad 2W_m = 92\pi > 48\pi$$

$$\therefore \text{Aliasing}$$

$$4\cos(46\pi t - k(2\pi(1/24)) + \pi/9)$$

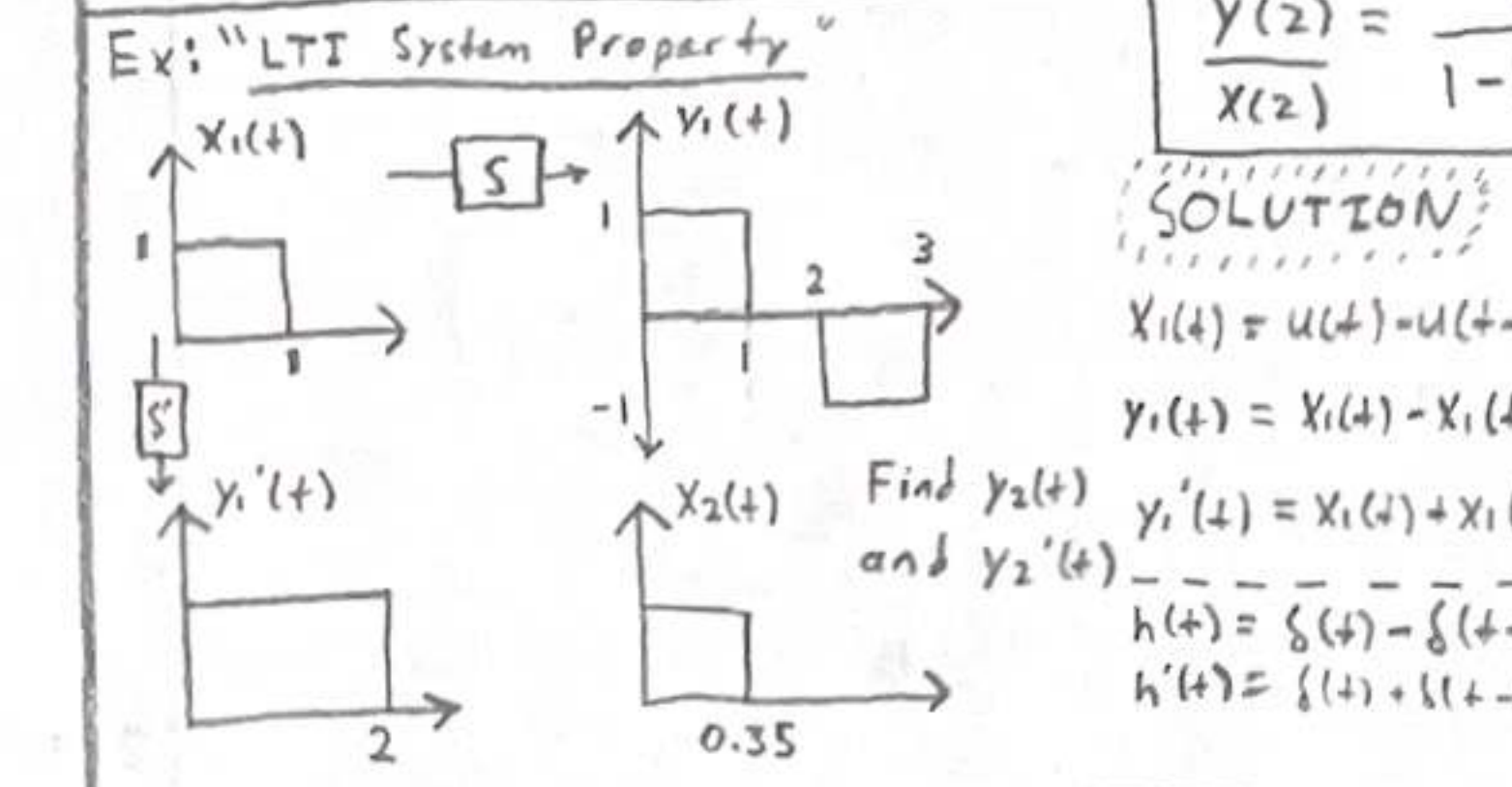
"choose integer value for k that puts it within nyquist range ie: 2"

$$4\cos(46\pi t - 48\pi t + \pi/9)$$

$$= 4\cos(-2\pi t + \pi/9)$$

$$\therefore w(t) = 7\cos(23\pi t) + 4\cos(-2\pi t + \pi/9)$$

Ex: "LTI System Property"



Find $y_2(t)$ and $y_2'(t)$

$$y_1(t) = x_1(t) - x_1(t-2)$$

$$y_1'(t) = x_1'(t) + x_1'(t-1)$$

$$h(t) = \delta(t) - \delta(t-2)$$

$$h'(t) = \delta'(t) + \delta'(t-1)$$

$$y_2(t) = x_2(t) * h(t) = x_2(t) - x_2(t-2)$$

$$y_2'(t) = x_2'(t) + x_2'(t-1)$$

$$y_2'(t) = \delta'(t) + \delta'(t-1)$$

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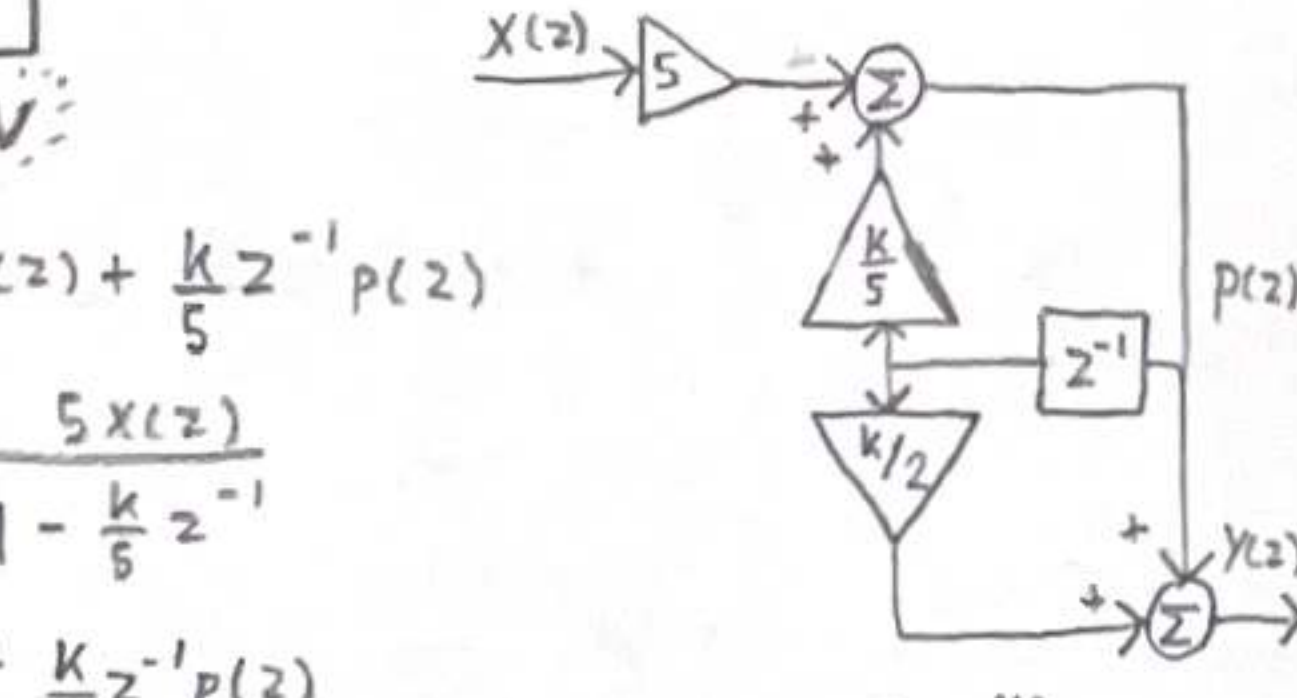
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Ex: "Block Diagram"

- Find the transfer function $H(z)$
- State the Radius of Convergence
- Find the values of $|k|$ for which the system is BIBO stable



$$p(z) = 5x(z) + \frac{k}{5}z^{-1}p(z)$$

$$\therefore p(z) = \frac{5x(z)}{1 - \frac{k}{5}z^{-1}}$$

$$Y(z) = p(z) + \frac{k}{2}z^{-1}p(z)$$

$$Y(z) = \frac{5x(z)}{1 - \frac{k}{5}z^{-1}} + \frac{k}{2} \cdot \frac{5x(z)z^{-1}}{1 - \frac{k}{5}z^{-1}}$$

$$\frac{Y(z)}{X(z)} = \frac{5}{1 - \frac{k}{5}z^{-1}} + \frac{k \cdot 5 \cdot z^{-1}}{2(1 - \frac{k}{5}z^{-1})}$$

$$H(z) = \frac{5(2 + \frac{k}{5})}{z - \frac{k}{5}}$$

$$\text{② ROC: } (\frac{k}{5}, \infty)$$

$$\text{③ BIBO: } (0, 5)$$

$$x_2(t) = u(t) - u(t-0.35)$$

$$\therefore y_2(t) = u(t) - u(t-0.35) - u(t-2) + u(t-0.35-2)$$

$$y_2'(t) = u'(t) - u'(t-0.35) + u'(t-1) - u'(t-0.35-1)$$

$$y_2'(t) = \delta(t) - \delta(t-0.35) + \delta(t-1) - \delta(t-1.35)$$

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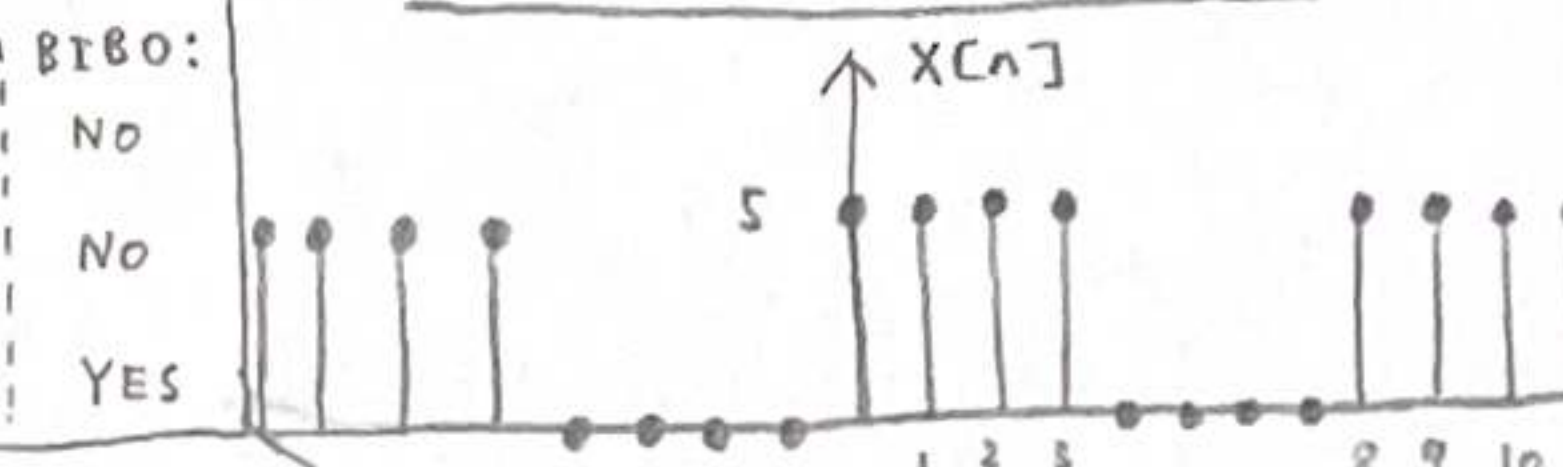
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Ex: "DT Fourier Series coefficients"



Find the fourier coefficients

$$X_k \text{ for } k \neq 0$$

SOLUTION:

$$N=8$$

$$W_0 = \frac{2\pi}{N} = \frac{\pi}{4}$$

$$X_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n]e^{-jk(\pi/4)n}$$

$$X_k = \frac{1}{8} \sum_{n=0}^7 (5)e^{-jk(\pi/4)n} + \emptyset$$

$$\text{known: } \sum_{n=0}^N ar^n = \frac{a(1-e^{(N+1)r})}{1-e^r}$$

$$a=5, r=e^{-jk(\pi/4)}, N=7$$

$$X_k = \frac{5(1-e^{-jk(\pi/4)8})}{1-e^{-jk(\pi/4)}} = 5 \cdot \frac{e^{-jk(\pi/4)4} [e^{jk(\pi/4)4} - 1]}{e^{-jk(\pi/4)4} [e^{jk(\pi/4)} - 1]}$$

$$= 5 \cdot \frac{e^{-jk(\pi/4)4} [e^{jk(\pi/4)4} - 1]}{e^{-jk(\pi/4)4} [e^{jk(\pi/4)} - 1]}$$

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Ex: "Diagonalization of a 3x3 Matrix"

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 2 & 3 \\ 3 & 3 & 20 \end{pmatrix} \rightarrow \det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 1 & 3 \\ 1 & 2-\lambda & 3 \\ 3 & 3 & 20-\lambda \end{vmatrix}$$

$$= (2-\lambda)[(2-\lambda)(20-\lambda)-9] - (1)[(20-\lambda)9] + (3)[3-3(2-\lambda)]$$

$$= \lambda^3 - 24\lambda^2 + 65\lambda - 42$$

$$\text{① Guess root } (\lambda-1)$$

$$= (\lambda-1)(\lambda^2 - 23\lambda + 42)$$

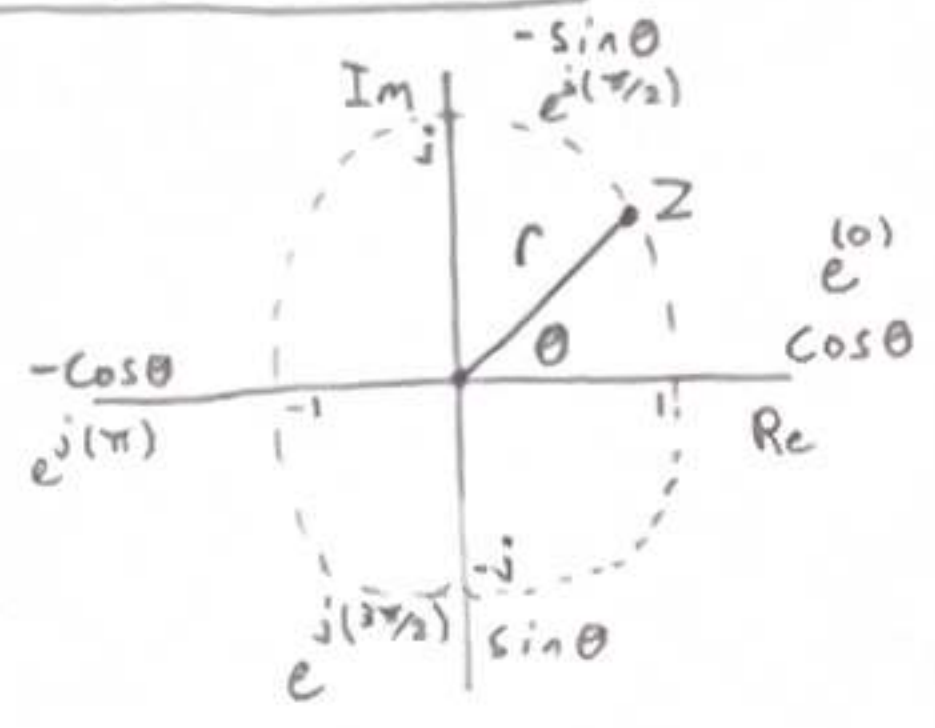
$$= (\lambda-1)(\lambda-2)(\lambda-21)$$

$$\text{For } \lambda=2: \text{ check to see if } \lambda=2 \text{ is a root}$$

$$A - 2I = \begin{pmatrix} 0 & 1 & 3 \\ 1 & 0 & 3 \\ 3 & 3 & 18 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Complex Numbers:



Cartesian Form:

$$Z = x + jy$$

Rectangular Form:

$$Z = (x, y)$$

Polar Form:

$$Z = r \angle \theta$$

Exponential Form:

$$Z = r e^{j\theta}$$

① Deterministic: VS. Stochastic:

Represented by a formula

Probabilistic Distribution

② Continuous: VS. Discrete:

Domain or range

Domain or range

③ Even:

VS. ODD:

$$x(t) = -x(-t)$$

⑤ Finite Ex/Px: VS. Infinite Ex/Px:

$$Ex < \infty$$
$$Px < \infty$$

$$Ex \rightarrow \infty$$
$$Px \rightarrow \infty$$

⑥ Causal:

$$x(t) = 0; t < 0$$

VS. Acausal:

Both sides of time (2 sided)

Euler Identity:

$$e^{j\theta} = \cos\theta + j \sin\theta$$

Even/Odd Signals

$$y(t) = y_e(t) + y_o(t)$$

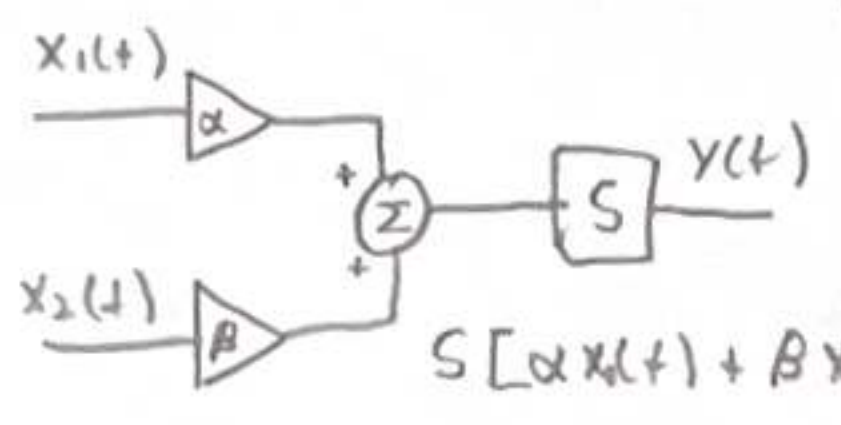
Energy / Power

$$Ex = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$Px = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

Systems

Linearity:



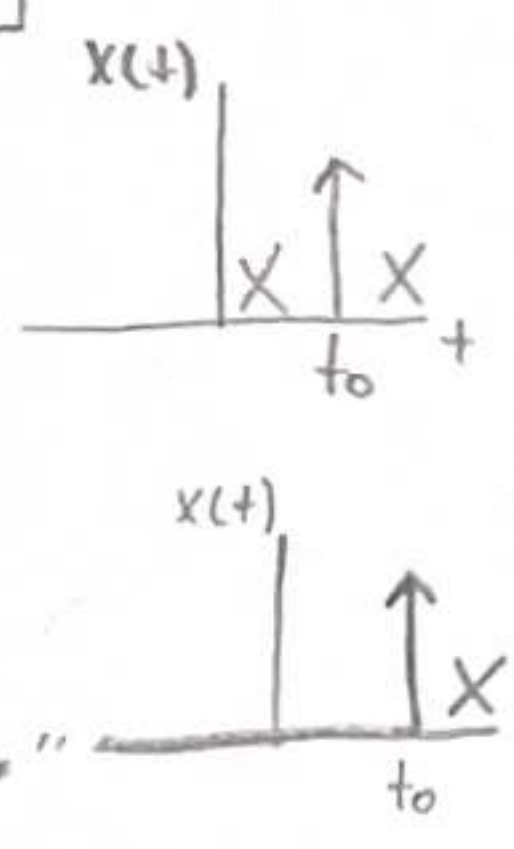
System Properties

① Memoryless:

"y(t) at t depends on x(t) at t"

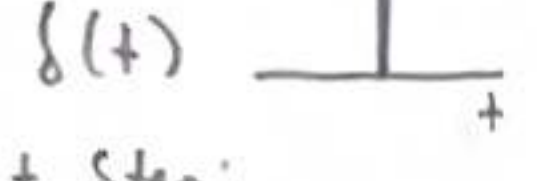
② Causality:

"y(t) at t depends on x(t) at t and the past but not future"

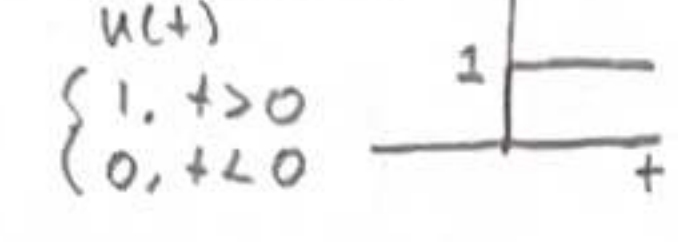


Basic Signals

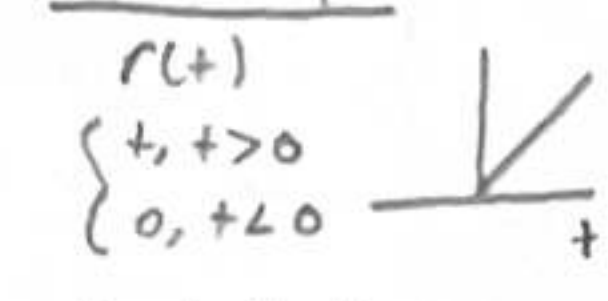
Dirac Delta:



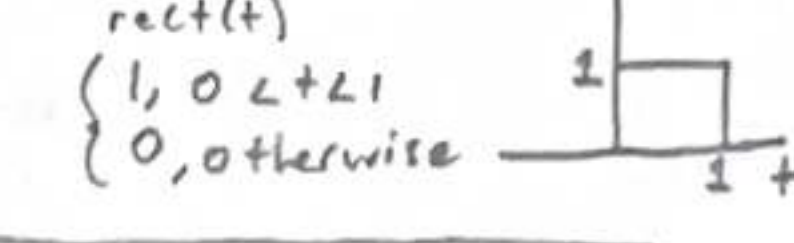
Unit Step:



Unit Ramp:



Rect function:



Convolution Integral

$$f(t) * g(t) = g(t) * f(t)$$

$$f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau) g(t-\tau) d\tau$$

Causal System

h(t) = 0, t < 0
"impulse response"

BIBO Stability

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty$$

Asymptotic Stability

Re(s) < 0 "Poles of transfer function"

Fourier Series

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_0 t}$$

$$X_k = \frac{1}{T} \int_0^T x(t) e^{-jk\omega_0 t} dt$$

Power (FS)

$$Px = \frac{1}{T} \int_0^T |x(t)|^2 dt$$

$$Px = \sum_{k=-\infty}^{\infty} |X_k|^2$$

Frequency Response

Eigenfunction Property, if the input is a complex exponential (or sinusoid), then yss(t) (steady state) is given by: (input x(t) = \alpha e^{j\omega t} + \cos(\omega t + \theta))

$$① y_{ss}(t) = |H(j\omega)| \alpha + |H(j\omega)| \cos(\omega t + \theta + \angle H(j\omega))$$

$$② x(t) = e^{j\omega t} : y_{ss}(t) = |H(j\omega)| e^{j(\omega t + \angle H(j\omega))}$$

$$\text{Re}(X_k) = C_k = \frac{1}{T} \int_0^T x(t) \cos(k\omega_0 t) dt$$

$$\text{Im}(X_k) = d_k = \frac{1}{T} \int_0^T x(t) \sin(k\omega_0 t) dt$$

$$|X_k| = \sqrt{C_k^2 + d_k^2}$$

$$\theta_k = -\tan^{-1}(\frac{d_k}{C_k})$$

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$$|X_k| = \sqrt{C_k^2 + d_k^2}$$

$$\theta_k = -\tan^{-1}(\frac{d_k}{C_k})$$

$$Re(X_k) = C_k = \frac{1}{T} \int_0^T x(t) \cos(k\omega_0 t) dt$$

$$\text{Im}(X_k) = d_k = \frac{1}{T} \int_0^T x(t) \sin(k\omega_0 t) dt$$

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$$\text{Im}(X_k) = d_k = \frac{1}{T} \int_0^T x(t) \sin(k\omega_0 t) dt$$

$$|X_k| = \sqrt{C_k^2 + d_k^2}$$

$$\theta_k = -\tan^{-1}(\frac{d_k}{C_k})$$

$$Re(X_k) = C_k = \frac{1}{T} \int_0^T x(t) \cos(k\omega_0 t) dt$$

Discrete time signals and systems

Convolution Sum:

Causality

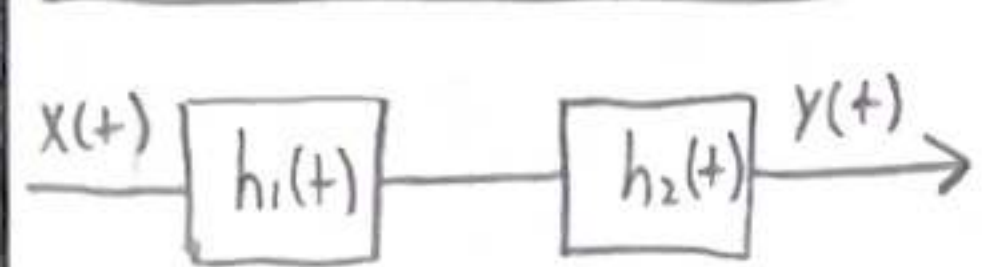
Causal LTI System:

Energy / Power:

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

$$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

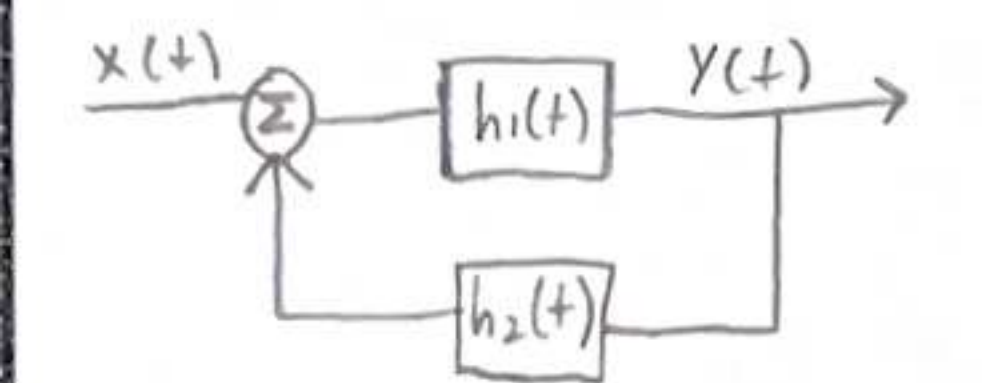
LTI System Connections



Cascade: $H(s) = H_1(s) \cdot H_2(s)$



Parallel: $H(s) = H_1(s) + H_2(s)$



Feedback: $H(s) = \frac{H_1(s)}{1 + H_2(s) \cdot H_1(s)}$

Controllability Matrix

$$x(t) = e^{At} x_0 + \int_0^t e^{A(t-\tau)} B u(\tau) d\tau$$

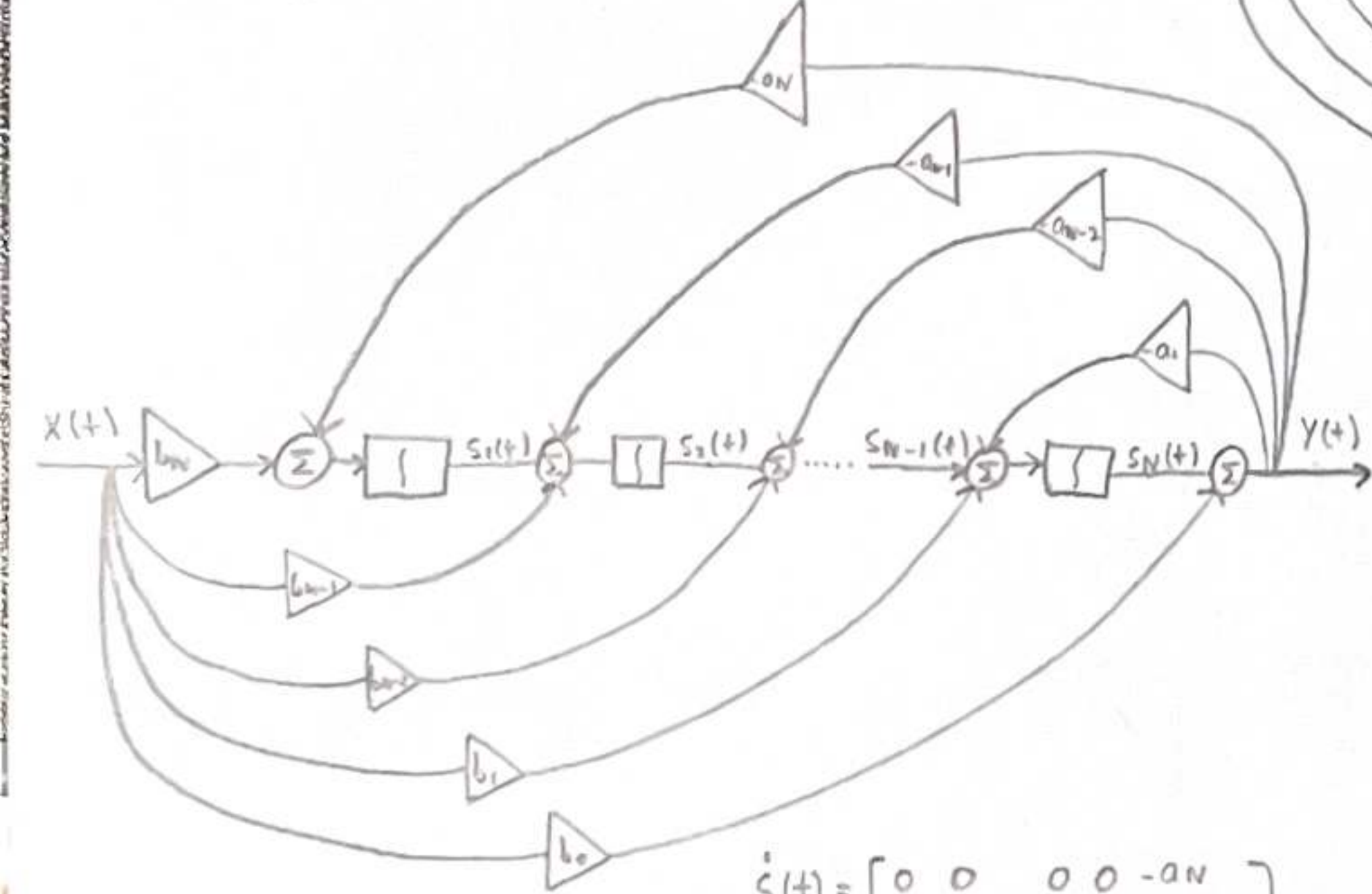
$$M_c = [B, AB, A^2B, \dots, A^{N-1}B]$$

$\det(M_c) \neq 0 \rightarrow$ Controllable

Observability Matrix

$$y(t) = C e^{At} x_0 + \int_0^t e^{A(t-\tau)} B u(\tau) d\tau + D u(\tau)$$

$$M_o = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{N-1} \end{bmatrix} \quad \det(M_o) \neq 0 \rightarrow \text{Observable}$$



SISO observable Canonical Form

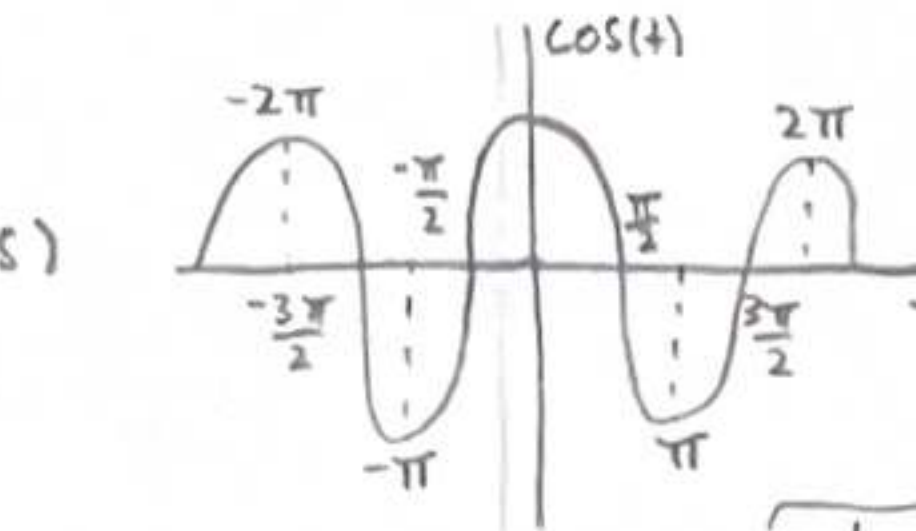
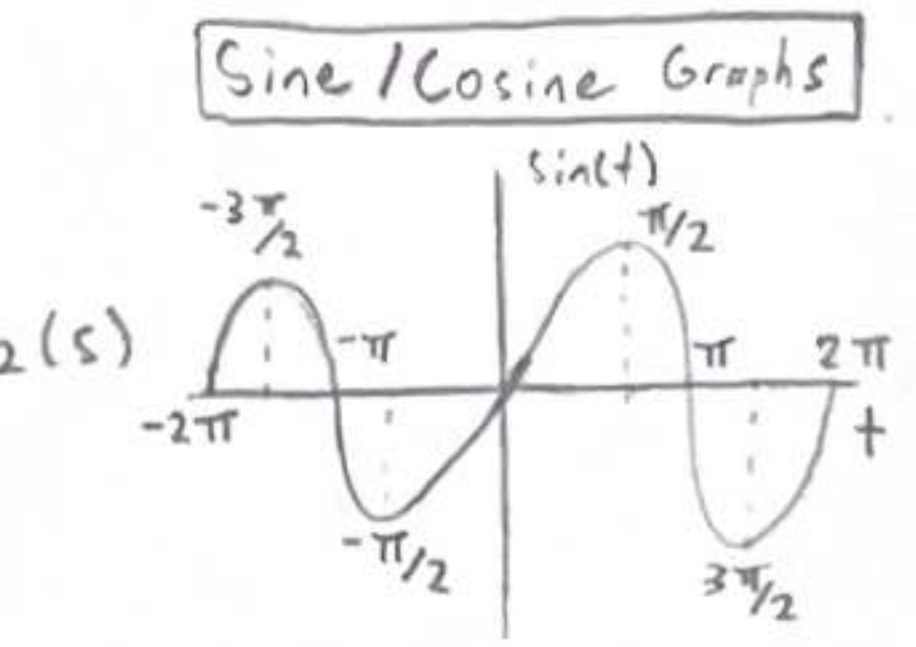
ODE: (same as CCF)

BIBO stability:

$$\sum_k |h[k]| < \infty$$

Asymptotic stability:

$|x| < 0$ "Poles of transfer function within unit circle"



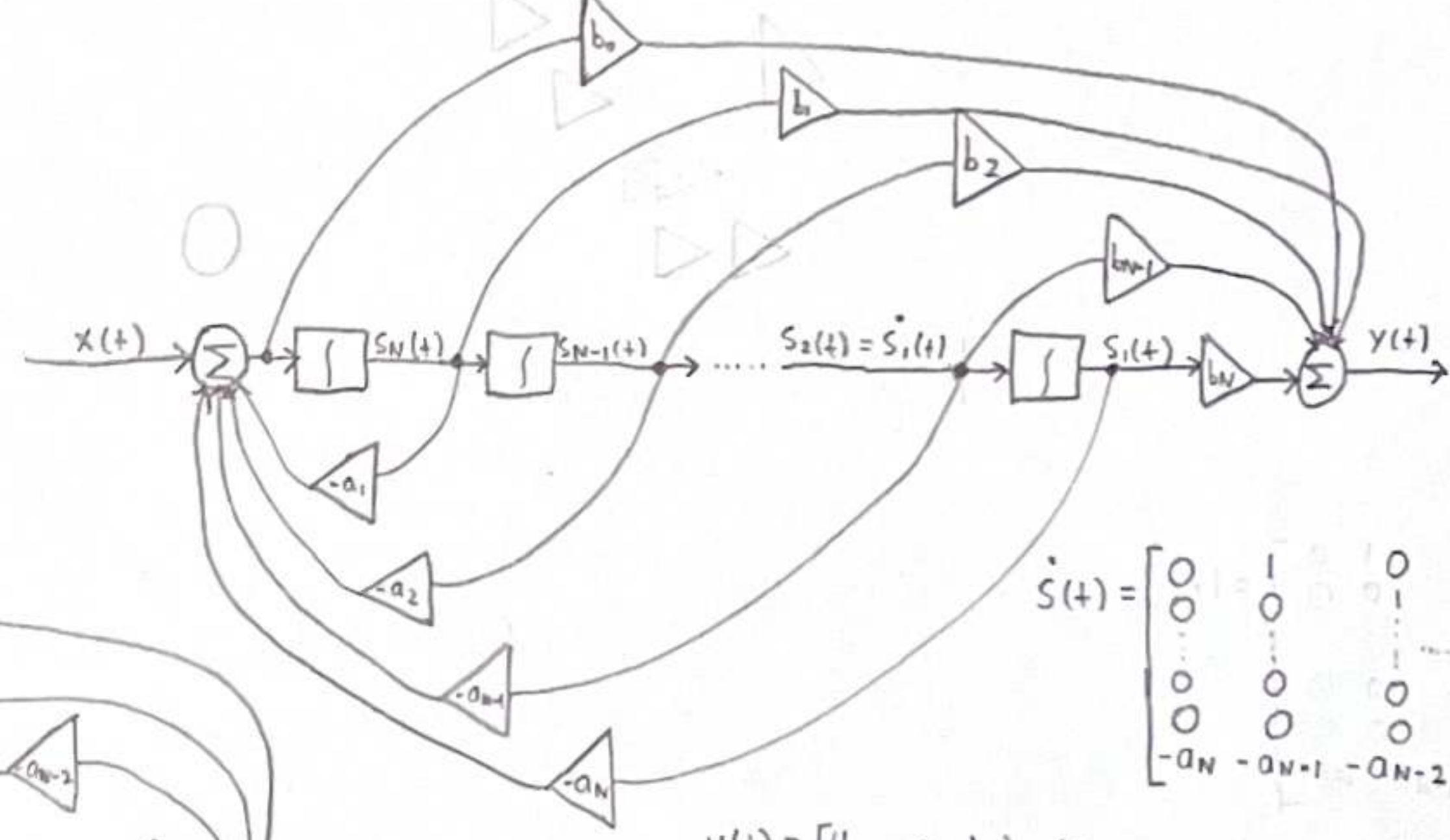
$$y(t) = \underbrace{C e^{At} x_0}_{\text{Zero-input response}} + \underbrace{C \int_0^t e^{A(t-\tau)} B u(\tau) d\tau + D u(t)}_{\text{Zero-state response}}$$

$$Y(s) = C(sI - A)^{-1} x_0 + [C(sI - A)^{-1} B + D] U(s)$$

$H(s) \rightarrow$ T.F Matrix

SISO Controllable Canonical Form

ODE: $\frac{d^N y(t)}{dt^N} + a_1 \frac{d^{N-1} y(t)}{dt^{N-1}} + a_2 \frac{d^{N-2} y(t)}{dt^{N-2}} + \dots + a_{N-1} \frac{dy(t)}{dt} + a_N y(t) = b_0 \frac{d^N x(t)}{dt^N} + b_1 \frac{d^{N-1} x(t)}{dt^{N-1}} + \dots + b_{N-1} \frac{dx(t)}{dt} + b_N x(t)$



$$\dot{s}(t) = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_N & -a_{N-1} & -a_{N-2} & \dots & -a_1 \end{bmatrix} s(t) + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} x(t)$$

$$y(t) = [(b_N - a_N b_0) \quad (b_{N-1} - a_{N-1} b_0) \quad \dots \quad (b_2 - a_2 b_0) \quad (b_1 - a_1 b_0)] s(t) + b_0 x(t)$$

Diagonalization of a matrix

$$A = T D T^{-1} \quad \text{where: } D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \text{ * eigenvalues of } A$$

$$D = T^{-1} A T$$

$$T = [v_1, v_2] \text{ * eigenvectors of } A$$

DT CCF/OCF

- ① $\{ \rightarrow z^{-1}$
- ② $x(t) \rightarrow x[n], y(t) \rightarrow y[n]$
- ③ $\dot{s}(t) = S[n+1]$

ODE: $y[n] + a_1 y[n-1] + \dots + a_N y[n-N] = b_0 x[n] + b_1 x[n-1] + \dots + b_N x[n-N]$

Table 4.1 Basic Properties of Fourier Series

Basic Properties of Fourier Series		
	Time Domain	Frequency Domain
Signals and constants	$x(t), y(t)$ periodic with period T_0, α, β	X_k, Y_k
Linearity	$\alpha x(t) + \beta y(t)$	$\alpha X_k + \beta Y_k$
Parseval's power relation	$P_x = \frac{1}{T_0} \int_{T_0} x(t) ^2 dt$	$P_x = \sum_k X_k ^2$
Differentiation	$\frac{dx(t)}{dt}$	$j k \Omega_0 X_k$
Integration	$\int_{-\infty}^t x(t') dt'$ only if $X_0 = 0$	$\frac{X_k}{j k \Omega_0}, k \neq 0$
Time shifting	$x(t - \alpha)$	$e^{-j k \Omega_0 \alpha} X_k$
Frequency shifting	$e^{j M \Omega_0 t} x(t)$	X_{k-M}
Symmetry	$x(t)$ real	$ X_k = X_{-k} $ even function of k $\angle X_k = -\angle X_{-k}$ odd function of k
Convolution in time	$z(t) = [x * y](t)$	$Z_k = X_k Y_k$

Table 3.1 Basic Properties of One-sided Laplace Transforms

	$\alpha f(t), \beta g(t)$	$\alpha F(s), \beta G(s)$
Causal functions and constants		
Linearity	$\alpha f(t) + \beta g(t)$	$\alpha F(s) + \beta G(s)$
Time shifting	$f(t - \alpha)u(t - \alpha)$	$e^{-\alpha s} F(s)$
Frequency shifting	$e^{at} f(t)$	$F(s - \alpha)$
Multiplication by t	$tf(t)$	$-\frac{dF(s)}{ds}$
Derivative	$\frac{df(t)}{dt}$	$sF(s) - f(0^-)$
Second derivative	$\frac{d^2 f(t)}{dt^2}$	$s^2 F(s) - sf(0^-) - f'(0^-)$
Integral	$\int_0^t f(t') dt'$	$\frac{F(s)}{s}$
Expansion/contraction	$f(\alpha t), \alpha \neq 0$	$\frac{1}{ \alpha } F\left(\frac{s}{\alpha}\right)$
Initial value	$f(0^-) = \lim_{s \rightarrow \infty} sF(s)$	

Table 3.2 One-sided Laplace Transforms

	Function of time	Function of s , ROC
(1)	$\delta(t)$	1, whole s -plane
(2)	$u(t)$	$\frac{1}{s}, \text{Re}[s] > 0$
(3)	$r(t)$	$\frac{1}{s^2}, \text{Re}[s] > 0$
(4)	$e^{-at} u(t), a > 0$	$\frac{1}{s+a}, \text{Re}[s] > -a$
(5)	$\cos(\Omega_0 t) u(t)$	$\frac{s}{s^2 + \Omega_0^2}, \text{Re}[s] > 0$
(6)	$\sin(\Omega_0 t) u(t)$	$\frac{\Omega_0}{s^2 + \Omega_0^2}, \text{Re}[s] > 0$
(7)	$e^{-at} \cos(\Omega_0 t) u(t), a > 0$	$\frac{s+a}{(s+a)^2 + \Omega_0^2}, \text{Re}[s] > -a$
(8)	$e^{-at} \sin(\Omega_0 t) u(t), a > 0$	$\frac{\Omega_0}{(s+a)^2 + \Omega_0^2}, \text{Re}[s] > -a$
(9)	$2Ae^{-at} \cos(\Omega_0 t + \theta) u(t), a > 0$	$\frac{A \angle \theta}{s+a-j\Omega_0} + \frac{A \angle -\theta}{s+a+j\Omega_0}, \text{Re}[s] > -a$
(10)	$\frac{1}{(N-1)!} t^{N-1} u(t)$	$\frac{1}{s^N}, N \text{ an integer}, \text{Re}[s] > 0$
(11)	$\frac{1}{(N-1)!} t^{N-1} e^{-at} u(t)$	$\frac{1}{(s+a)^N}, N \text{ an integer}, \text{Re}[s] > -a$
(12)	$\frac{2A}{(N-1)!} t^{N-1} e^{-at} \cos(\Omega_0 t + \theta) u(t)$	$\frac{A \angle \theta}{(s+a-j\Omega_0)^N} + \frac{A \angle -\theta}{(s+a+j\Omega_0)^N}, \text{Re}[s] > -a$

No.	$x_1(t)$	$x_2(t)$	$x_1(t) * x_2(t) = x_2(t) * x_1(t)$
1	$x(t)$	$\delta(t - T)$	$x(t - T)$
2	$e^{\lambda_1 t} u(t)$	$u(t)$	$\frac{1 - e^{\lambda_1 t}}{-\lambda_1} u(t)$
3	$u(t)$	$u(t)$	$tu(t)$
4	$e^{\lambda_1 t} u(t)$	$e^{\lambda_2 t} u(t)$	$\frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_1 - \lambda_2} u(t) \quad \lambda_1 \neq \lambda_2$
5	$e^{\lambda_1 t} u(t)$	$e^{\lambda_2 t} u(t)$	$te^{\lambda_2 t} u(t)$
6	$te^{\lambda_1 t} u(t)$	$e^{\lambda_2 t} u(t)$	$\frac{1}{2} t^2 e^{\lambda_2 t} u(t)$

L2.4 p17

Table 5.1 Basic Properties of Fourier Transform

	Time Domain	Frequency Domain
Signals and constants	$x(t), y(t), z(t), \alpha, \beta$	$X(\Omega), Y(\Omega), Z(\Omega)$
Linearity	$\alpha x(t) + \beta y(t)$	$\alpha X(\Omega) + \beta Y(\Omega)$
Expansion/contraction in time	$x(\alpha t), \alpha \neq 0$	$\frac{1}{ \alpha } X\left(\frac{\Omega}{\alpha}\right)$
Reflection	$x(-t)$	$X(-\Omega)$
Parseval's energy relation	$E_x = \int_{-\infty}^{\infty} x(t) ^2 dt$	$E_x = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) ^2 d\Omega$
Duality	$X(t)$	$2\pi x(-\Omega)$
Time differentiation	$\frac{d^n x(t)}{dt^n}, n \geq 1, \text{ integer}$	$(j\Omega)^n X(\Omega)$
Frequency differentiation	$-\frac{d}{dt} x(t)$	$\frac{dX(\Omega)}{d\Omega}$
Integration	$\int_{-\infty}^t x(t') dt'$	$\frac{X(\Omega)}{j\Omega} + \pi X(0) \delta(\Omega)$
Time shifting	$x(t - \alpha)$	$e^{-j\alpha\Omega} X(\Omega)$
Frequency shifting	$e^{j\Omega_0 t} x(t)$	$X(\Omega - \Omega_0)$
Modulation	$x(t) \cos(\Omega_c t)$	$0.5[X(\Omega - \Omega_c) + X(\Omega + \Omega_c)]$
Periodic signals	$x(t) = \sum_k X_k e^{jk\Omega_0 t}$	$X(\Omega) = \sum_k 2\pi X_k \delta(\Omega - k\Omega_0)$
Symmetry	$x(t)$ real	$ X(\Omega) = X(-\Omega) $ $\angle X(\Omega) = -\angle X(-\Omega)$
Convolution in time	$z(t) = [x * y](t)$	$Z(\Omega) = X(\Omega)Y(\Omega)$
Windowing/Multiplication	$x(t)y(t)$	$\frac{1}{2\pi} [X * Y](\Omega)$
Cosine transform	$x(t)$ even	$X(\Omega) = \int_{-\infty}^{\infty} x(t) \cos(\Omega t) dt, \text{ real}$
Sine transform	$x(t)$ odd	$X(\Omega) = -j \int_{-\infty}^{\infty} x(t) \sin(\Omega t) dt, \text{ imaginary}$

Table 5.2 Fourier Transform Pairs

	Function of Time	Function of Ω
(1)	$\delta(t)$	1
(2)	$\delta(t - \tau)$	$e^{-j\Omega\tau}$
(3)	$u(t)$	$\frac{1}{j\Omega} + \pi\delta(\Omega)$
(4)	$u(-t)$	$\frac{-1}{j\Omega} + \pi\delta(\Omega)$
(5)	$\text{sign}(t) = 2[u(t) - 0.5]$	$\frac{2}{j\Omega}$
(6)	$A, -\infty < t < \infty$	$2\pi A\delta(\Omega)$
(7)	$Ae^{-at} u(t), a > 0$	$\frac{A}{j\Omega + a}$
(8)	$Ate^{-at} u(t), a > 0$	$\frac{A}{(j\Omega + a)^2}$
(9)	$e^{-a t }, a > 0$	$\frac{2a}{a^2 + \Omega^2}$
(10)	$\cos(\Omega_0 t), -\infty < t < \infty$	$\pi[\delta(\Omega - \Omega_0) + \delta(\Omega + \Omega_0)]$
(11)	$\sin(\Omega_0 t), -\infty < t < \infty$	$-j\pi[\delta(\Omega - \Omega_0) - \delta(\Omega + \Omega_0)]$
(12)	$\rho(t) = A[u(t + \tau) - u(t - \tau)], \tau > 0$	$2A\tau \frac{\sin(\Omega\tau)}{\Omega\tau}$
(13)	$\frac{\sin(\Omega_0 t)}{\pi t}$	$P(\Omega) = u(\Omega + \Omega_0) - u(\Omega - \Omega_0)$
(14)	$x(t) \cos(\Omega_0 t)$	$0.5[X(\Omega - \Omega_0) + X(\Omega + \Omega_0)]$

Fourier Series of Discrete-time Periodic signals

	Time Domain	Frequency Domain
	$x[n]$ periodic signal of period N	$X[k]$ periodic FS coefficients of period N
Z-transform	$X_1(z) = \sum_n x[n] z^{-n}$	$X[k] = \frac{1}{N} \sum_n X_1(e^{j\omega}) _{\omega = 2\pi k/N}$
DTFT	$X(e^{j\omega}) = \sum_k X[k] e^{j\omega k}$	$X(e^{j\omega}) = \sum_k 2\pi X[k] \delta(\omega - 2\pi k/N)$
LTI response	input $x[n] = \sum_k X[k] e^{j\omega_k n}$	output: $y[n] = \sum_k X[k] H(e^{j\omega_k}) e^{j\omega_k n}$ $H(e^{j\omega})$ (frequency response of system)
Time-shift (circular shift)	$x[n - M]$	$X[k] e^{-j2\pi kM/N}$
Modulation	$x[n] e^{j2\pi Mn/N}$	$X[k - M]$
Multiplication	$x[n] y[n]$	$\sum_{m=0}^{N-1} X[m] Y[k - m]$ periodic convolution
Periodic convolution	$\sum_{m=0}^{N-1} x[m] y[n - m]$	$NX[k] Y[n]$

Properties of the DTFT

Z-transform:	$x[n], X(z), z = 1 \in \text{ROC}$	$X(e^{j\omega}) = X(z) _{z=e^{j\omega}}$
Periodicity:	$x[n]$	$X(e^{j\omega}) = X(e^{j(\omega + 2\pi k)}), k \text{ integer}$
Linearity:	$\alpha x[n] + \beta y[n]$	$\alpha X(e^{j\omega}) + \beta Y(e^{j\omega})$
Time-shifting:	$x[n - N]$	$e^{-j\omega N} X(e^{j\omega})$
Frequency-shift:	$x[n] e^{j\omega_0 n}$	$X(e^{j(\omega - \omega_0)})$
Convolution:	$(x * y)[n]$	$X(e^{j\omega}) Y(e^{j\omega})$
Multiplication:	$x[n] y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta}) Y(e^{j(\omega - \theta)}) d\theta$
Symmetry:	$x[n]$ real-valued	$ X(e^{j\omega}) $ even function of ω $\angle X(e^{j\omega})$ odd function of ω
Parseval's relation:	$\sum_{n=-\infty}^{\infty} x[n] ^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) ^2 d\omega$	

One-sided Z-transforms

Function of Time	Function of z, ROC
(1) $\delta[n]$	1, Whole z-plane
(2) $u[n]$	$\frac{1}{1-z^{-1}}, z > 1$
(3) $nu[n]$	$\frac{z^{-1}}{(1-z^{-1})^2}, z > 1$
(4) $n^2 u[n]$	$\frac{z^{-1}(1+z^{-1})}{(1-z^{-1})^3}, z > 1$
(5) $\alpha^n u[n], \alpha < 1$	$\frac{1}{1-\alpha z^{-1}}, z > \alpha $
(6) $n\alpha^n u[n], \alpha < 1$	$\frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}, z > \alpha $
(7) $\cos(\omega_0 n) u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}, z > 1$
(8) $\sin(\omega_0 n) u[n]$	$\frac{\sin(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}, z > 1$
(9) $\alpha^n \cos(\omega_0 n) u[n], \alpha < 1$	$\frac{1-\alpha \cos(\omega_0)z^{-1}}{1-2\alpha \cos(\omega_0)z^{-1}+\alpha^2 z^{-2}}, z > 1$
(10) $\alpha^n \sin(\omega_0 n) u[n], \alpha < 1$	$\frac{\alpha \sin(\omega_0)z^{-1}}{1-2\alpha \cos(\omega_0)z^{-1}+\alpha^2 z^{-2}}, z > \alpha $

Table 10.2 Basic Properties of One-sided Z-transform

Causal signals and constants	$\alpha x[n], \beta y[n]$	$\alpha X(z), \beta Y(z)$
Linearity	$\alpha x[n] + \beta y[n]$	$\alpha X(z) + \beta Y(z)$
Convolution sum	$(x * y)[n] = \sum_k x[k]y[n-k]$	$X(z)Y(z)$
Time shifting - causal	$x[n-N], N \text{ integer}$	$z^{-N}X(z)$
Time shifting - non-causal	$x[n-N]$	$z^{-N}X(z) + x[-1]z^{-N+1}$
	$x[n] \text{ non-causal}, N \text{ integer}$	$+x[-2]z^{-N+2} + \dots + x[-N]$
Time reversal	$x[-n]$	$X(z^{-1})$
Multiplication by n	$nx[n]$	$-z \frac{dX(z)}{dz}$
Multiplication by n^2	$n^2 x[n]$	$z^2 \frac{d^2 X(z)}{dz^2} + z \frac{dX(z)}{dz}$
Finite difference	$x[n] - x[n-1]$	$(1-z^{-1})X(z) - x[-1]$
Accumulation	$\sum_{k=0}^n x[k]$	$\frac{X(z)}{1-z^{-1}}$
Initial value	$x[0]$	$\lim_{z \rightarrow \infty} X(z)$
Final value	$\lim_{n \rightarrow \infty} x[n]$	$\lim_{z \rightarrow 1} (z-1)X(z)$

Discrete-time Fourier Transforms (DTFT)

Discrete-time signal	DTFT $X(e^{j\omega})$, periodic of period 2π
(1) $\delta[n]$	$1, -\pi \leq \omega < \pi$
(2) $A \alpha^n u[n-1] \leftrightarrow \frac{e^{-j\omega}}{1-\alpha e^{-j\omega}}$	$2\pi A \delta(\omega), -\pi \leq \omega < \pi$
(3) $e^{j\omega_0 n}$	$2\pi \delta(\omega - \omega_0), -\pi \leq \omega < \pi$
(4) $\alpha^n u[n], \alpha < 1$	$\frac{1}{1-\alpha e^{-j\omega}}, -\pi \leq \omega < \pi$
(5) $n \alpha^n u[n], \alpha < 1$	$\frac{\alpha e^{-j\omega}}{(1-\alpha e^{-j\omega})^2}, -\pi \leq \omega < \pi$
(6) $\cos(\omega_0 n) u[n]$	$\pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)], -\pi \leq \omega < \pi$
(7) $\sin(\omega_0 n) u[n]$	$-j\pi [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)], -\pi \leq \omega < \pi$
(8) $\alpha^n, \alpha < 1$	$\frac{1-\alpha^2}{1-2\alpha \cos(\omega) + \alpha^2}, -\pi \leq \omega < \pi$
(9) $p[n] = u[n+N/2] - u[n-N/2]$	$\frac{\sin(\omega(N+1)/2)}{\sin(\omega/2)}, -\pi \leq \omega < \pi$
(10) $\alpha^n \cos(\omega_0 n) u[n]$	$\frac{1-\alpha \cos(\omega_0) e^{-j\omega}}{1-2\alpha \cos(\omega_0) e^{-j\omega} + \alpha^2 e^{-2j\omega}}, -\pi \leq \omega < \pi$
(11) $\alpha^n \sin(\omega_0 n) u[n]$	$\frac{\alpha \sin(\omega_0) e^{-j\omega}}{1-2\alpha \cos(\omega_0) e^{-j\omega} + \alpha^2 e^{-2j\omega}}, -\pi \leq \omega < \pi$

Discrete Fourier Transform (DFT) (Fourier Series Coefficients)

$x[n]$ finite-length N aperiodic signal	$\tilde{x}[n]$ periodic extension of period $L \geq N$
$\tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{L-1} \tilde{x}[k] e^{j2\pi nk/L}$	$\tilde{X}[k] = \sum_{n=0}^{L-1} \tilde{x}[n] e^{-j2\pi nk/L}$
IDFT/DFT $x[n] = \tilde{x}[n]W[N], W[n] = u[n] - u[n-N]$	$X[k] = \tilde{X}[k]W[k], W[k] = u[k] - u[k-N]$
Circular convolution $(x \otimes_L y)[n]$	$X[k]Y[k]$
Circular and linear convolution $(x \otimes_L y)[n] = (x * y)[n], L \geq M+K-1$ $M = \text{length of } x[n], K = \text{length of } y[n]$	$\sum_{n=0}^N a r^n = \frac{a(1-r^{N+1})}{1-r}$

Euler's formula

$$e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

... for cosine

$$\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

... for sine

$$\sin(\theta) = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

sinc function

$$\text{sinc}(\theta) := \frac{\sin(\pi\theta)}{\pi\theta}$$

Recap of Transforms

$$\text{LT: } X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

$$\text{ILT: } x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(s) e^{st} ds$$

$$\text{FT: } X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$\text{IFT: } x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$\text{FS: } X_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt$$

$$\text{IFS: } x(t) = \sum_{k=-\infty}^{\infty} X_k e^{jk\omega_0 t}$$

$$\text{ZT: } X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$\text{IZT: } x[n] = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz$$

$$\text{DTFT: } X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$\text{IDTFT: } x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$$

$$\text{DFT: } X_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk\omega_0 n}$$

$$\text{IDFT: } x[n] = \sum_{k=0}^{N-1} X_k e^{jk\omega_0 n}$$

$$\text{Where } \omega_0 = \frac{2\pi}{N}$$

3. Sum-Difference Formulas

$$\sin(x+y) = \sin x \cos y + \cos x \sin y$$

$$\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$\sin(x-y) = \sin x \cos y - \cos x \sin y$$

$$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$

$$\cos(x-y) = \cos x \cos y + \sin x \sin y$$

7. Double Angle Formulas

$$\sin(2x) = 2 \sin x \cos x$$

$$\cos(2x) = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$\tan(2x) = \frac{2 \tan x}{1 - \tan^2 x}$$

3. Power-Reducing/Half Angle Formulas

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

$$\tan^2 x = \frac{1 - \cos 2x}{1 + \cos 2x}$$

$$\sin \frac{x}{2} = \pm \sqrt{\frac{1 - \cos x}{2}}$$

$$\cos \frac{x}{2} = \pm \sqrt{\frac{1 + \cos x}{2}}$$

$$\tan \frac{x}{2} = \pm \sqrt{\frac{1 - \cos x}{1 + \cos x}} = \frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}$$